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Market Study

Vopak

Intermediary Report

21 November 2025

AGENDA

- **Current market overview**
- Demand analysis
  - Diesel
  - Gasoline
- Demand forecast
- Supply dynamics & forecast
- Namibia-Walvis Bay overview
- Competitor Analysis – Vitol
- Biofuels Blending Regulations

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Phase 1 is progressing well, with key elements being refined as the project advances into Phase 2.

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	Kick-off	Week 1 (6 Oct – 10 Oct)	Week 2 (13 Oct – 17 Oct)	Week 3 (20 Oct – 24 Oct)	Week 4 (27 Oct – 31 Oct)	Week 5 (3 Nov – 7 Nov)	Week 6 (10 Nov – 14 Nov)
Project scope alignment	✓	✓	✓	✓			
Phase 1 – Situational Analysis, Information Gathering in Long Term Demand and Supply Dynamics	• Desktop Research, data gathering and analysis • Consolidation of Desktop Data into Insights • Liquid fuels model updates (Diesel, Gasoline)						
Phase 2 – Competitor Analysis and Southern African Gas Landscape					• Development of competitor outlook and investigate Vitol activities in South Africa • High level overview of Walvis Bay opportunities analysing customs data in SADC region. • Biofuels Blending regulations • LNG/Gas & LPG		
Phase 3 – Hand-over & Board Presentation						• Development of Recommendations Report • Development of Hand-over presentation	

Current Progress

Black	Upcoming	Green	Complete	Orange	In Progress

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Following the first progress update, further data gathering has done towards demand and supply modelling.

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Next Steps: Update

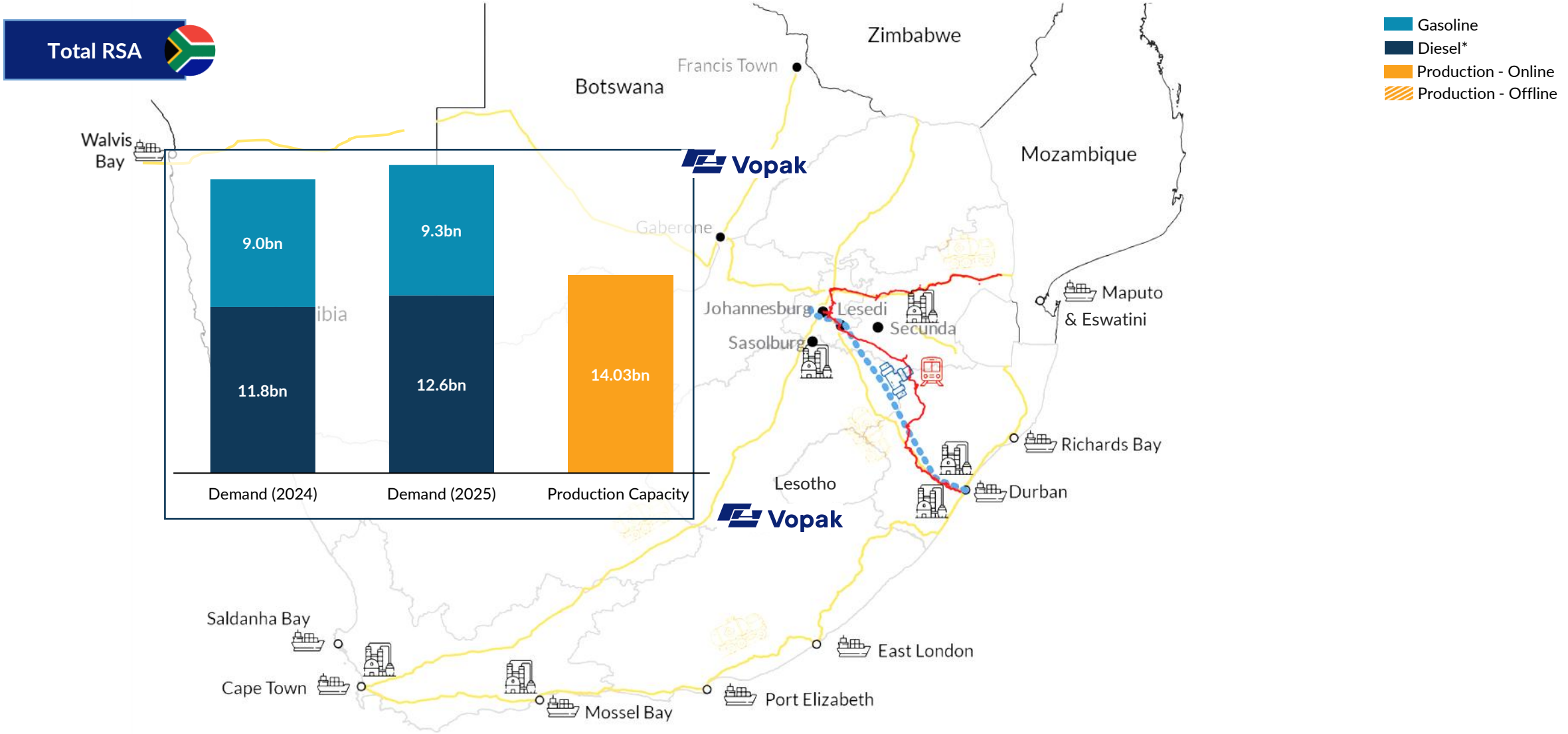
Demand side modelling

- Characterize and quantify vehicle efficiency gains
- Analyze and quantify latest vehicle mix trends (Traditional hybrids, plug-in hybrids, EV's, petrol and diesel)

Supply side modelling

- Develop a high-level carbon balance model for Secunda to assess impacts on liquid fuels production.
- Quantify production changes from 2025–2028 following destoning operations.
- Evaluate the effects of MRG cutbacks and LNG integration on output from 2028–2030.
- Project production changes post-2030, including LNG start-up and the 2032 gas cliff.

The South African demand for liquid fuels has increased but production capacity remains below demand…



Source: Figures in billions and rounded off, SARS Tariff data 2024 & 2025 Annualized; *Vopak market share in the coastal province is KZN*

Inland supply is likely to improve with improved Secunda supply & Natref c, SAPREF restart remains highly unlikely …

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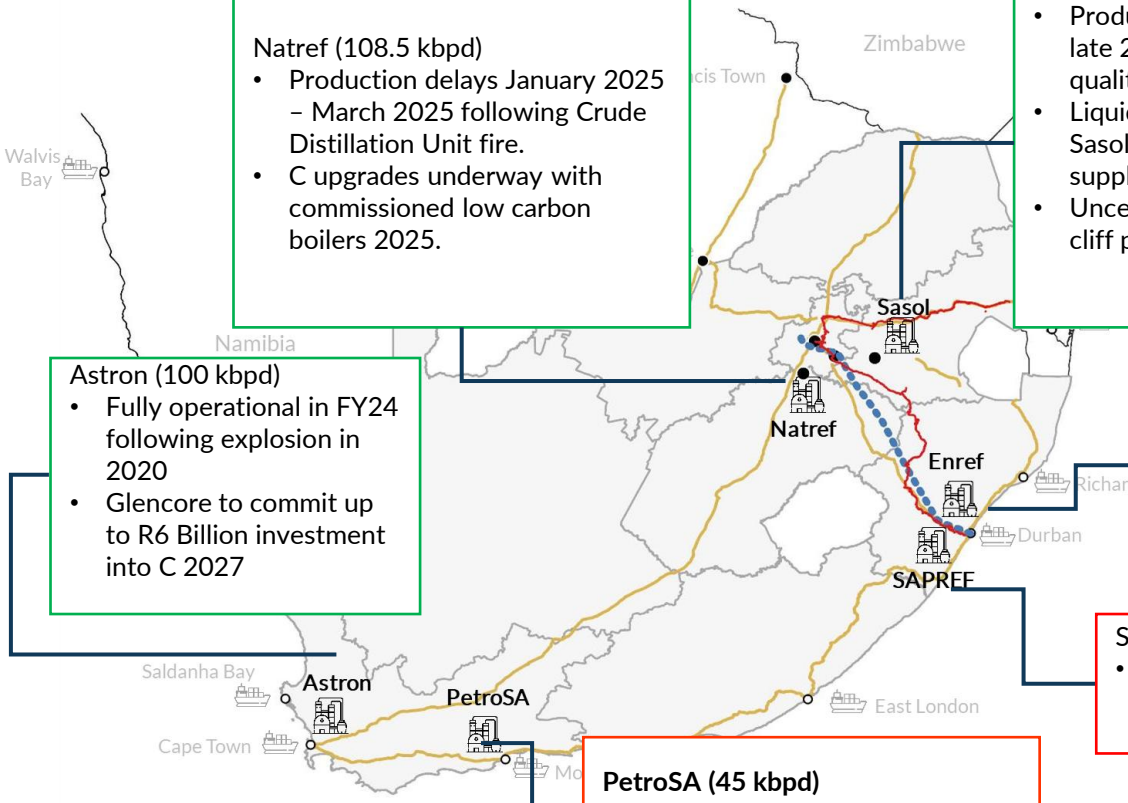
Local supply landscape

HP – High production; LP – Low production; HI – High import; LI – Low import

Natref (108.5 kbpd)
- Production delays January 2025 – March 2025 following Crude Distillation Unit fire.
- C upgrades underway with commissioned low carbon boilers 2025.

Secunda (75 kbpd)
- Production is expected to improve in late 2025-2028 due to improved coal quality through destoning.
- Liquid fuels production at risk with Sasol commitment to maintain supply of MRG through 2028 – 2030
- Uncertainty regarding looming gas cliff post 2032



Astron (100 kbpd)
- Fully operational in FY24 following explosion in 2020
- Glencore to commit up to R6 Billion investment into C 2027

Enref (120 kbpd)
- Shutdown in 2021 following explosion/fire

SAPREF (180 kbpd)
- Shutdown in 2022 following extensive flood damage. Acquired by CEF in 2024

PetroSA (45 kbpd)
- HP - Planned to restart in 2029/30 due to Total exit of 11B/12B fields
- LP – does not restart

Source: The African Report; Reuters; Africa Oil and Gas Report; AIA Analysis

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...leading to an increased reliance on imports in 2025

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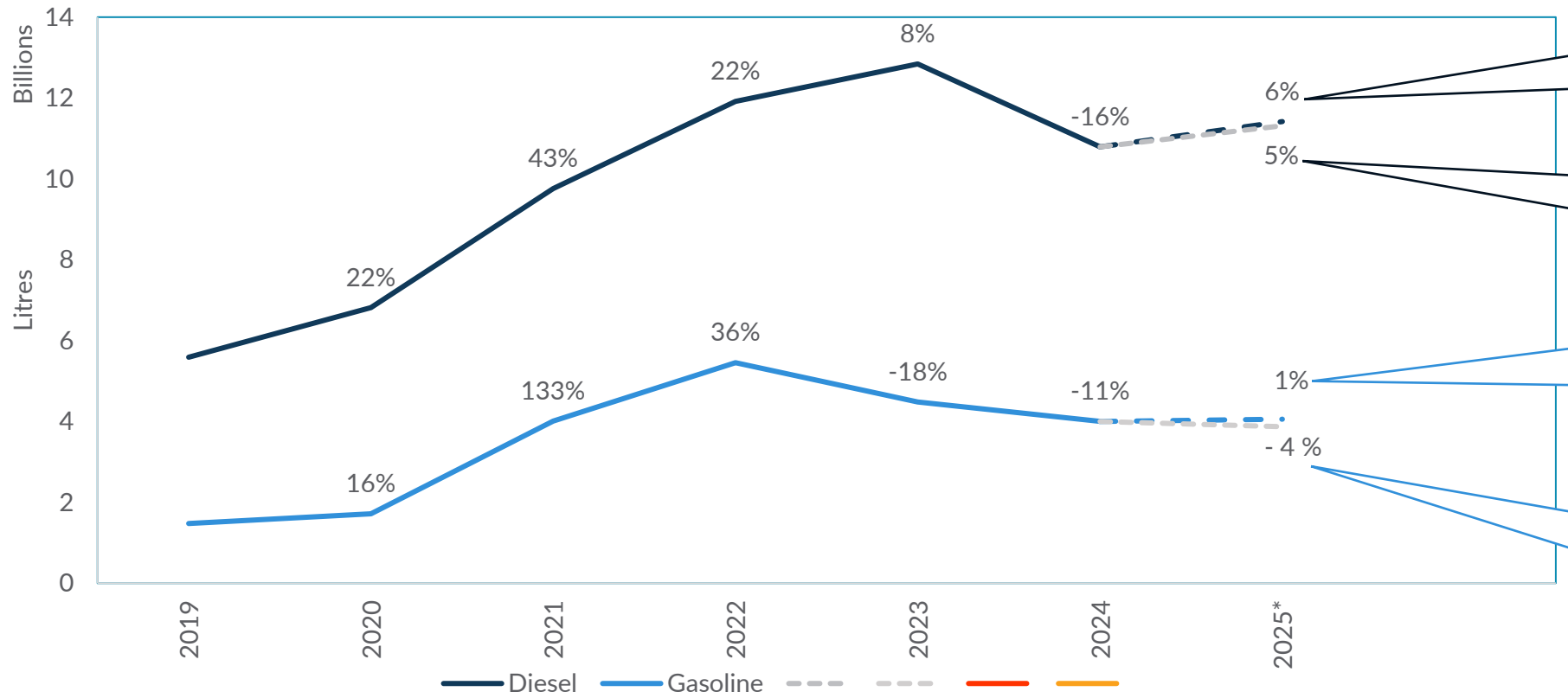
Diesel and gasoline imports and year-on-year change



Fire at Sasol's Natref facility resulted in loss of two months production that was met by imports.

Change in diesel imports if there was no loss of production at Natref

Falling gasoline prices has resulted in a slight pick up in gasoline demand between 2024 and 2025, while Secunda has experienced lower production between FY22-25

Change in gasoline imports if there was no loss of production at Natref

Reliance on imports is growing in 2025 due to an unstable supply landscape

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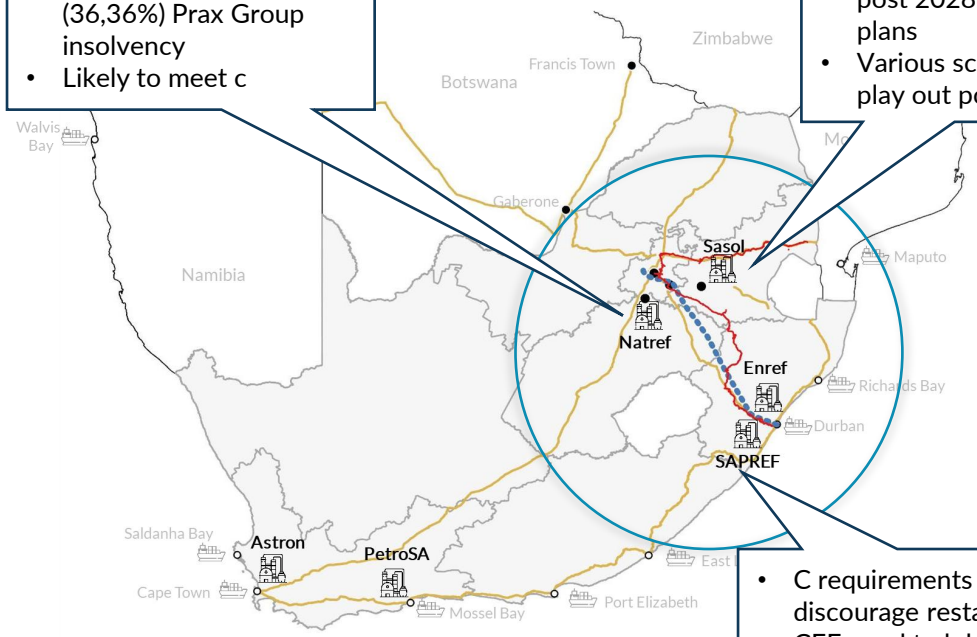
Source: Figures in billions, SARS Fuel import, 2019-25; 2025 is annualised, StatsSA*

Several characteristics influence the projections for liquid fuels demand and supply

Supply forecast considerations

- Natref to continue at current operations despite joint owner (36,36%) Prax Group insolvency
- Likely to meet c

- Destoning project will see steady improvement in production post December 2025
- Cut back in production post 2028 due to MRG plans
- Various scenarios can play out post 2030



- C requirements discourage restart
- CEF need to lobby for funds to restart refinery with no financing details available.
- Disruptions from environmentalists due to pollution

Demand forecast drivers

Diesel

- ↓ Improved coal fleet performance (EAF ↑ to 62.8%) lessened the need for diesel for generation, but diesel peakers to shift from crisis supply to renewable integration support. Although 2025 has seen a temporary increase in diesel peaker usage
- ↓ Slight improvements in rail freight point to a decrease in diesel demand as more payload shifts from road to rail.
- ↑ When SA experiences economic growth (potentially increase in GDP per capita) several diesel heavy industries drive demand such as construction and mining.

Gasoline

- ↑ A reduction in fuel price driven by lowering oil prices, a strengthening rand currency and improved local production drive up gasoline demand
- ↓ EV sales have increased by 22% in 2024 compared to the previous year, reducing the need for gasoline-powered vehicles
- ↓ Traditional hybrid vehicle sales saw 70,9% year on year in Q1 of 2025, forming the largest part of NEV's and with potential to reduce demand for gasoline.

Vopak has several opportunities for growth and areas of concern, to be addressed, to drive business expansion and mitigate potential risks

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Potential opportunities

- 40 000 m³ of storage in Durban presents an opportunity for Vopak to target players like Vitol through partnerships that enhance fuel supply security and market growth. Additionally, Durban can act as a break bulk facility to coastal airports.
- Lower gasoline prices and reduced local supply from Sasol's Secunda (MRG cutback) present an opportunity to increase storage demand for gasoline imports
- Engaging Transnet on the NMPP pricing model under Section 79 offers Vopak a chance to influence tariff structures, enhance the economics of coastal storage, and strengthen its strategic role in South Africa's fuel logistics system.

Concerns

- Strategic measures like tailored time-of-use tariffs and optimized charging station deployment with increased embedded and private solar PV could drive greater EV adoption – implying a lower demand for gasoline
- Given the coal fleet's capacity factor remains around 63%, diesel demand for power generation decreases – impacting diesel consumption for OCGTs.
- Potential revival of rail transport in SA resulting in a switch of goods transport from road to rail

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The forecasts for liquid fuels can be explained by both cyclical as well as structural factors

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Key Commodity Demand Drivers

Diesel

GDP	MTBPS Low:1.0% MTBPS High 1.6%
Vehicle efficiency	Low: 0.5% High 1.0%
Power generation	Eskom forecasts and AIA modelling
Shift between rail and road	Operation Vulindlela outlook & industry view

Gasoline

Price	Based on current prices and CPI index
Vehicle efficiency	Low: 0.5% High 1.0%
Vehicle mix	Statistic South Africa, Vehicle classification YoY change
GDP / capita	UNDP Low: 0.5% UNDP High 0.01%
Kilometers travelled	Based on vehicle changing travel pattens

Assumptions based on 2025 growth rates

Cyclical | Structural

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Diesel demand is closely linked to GDP but has occasionally exceeded GDP growth due to structural changes in the market

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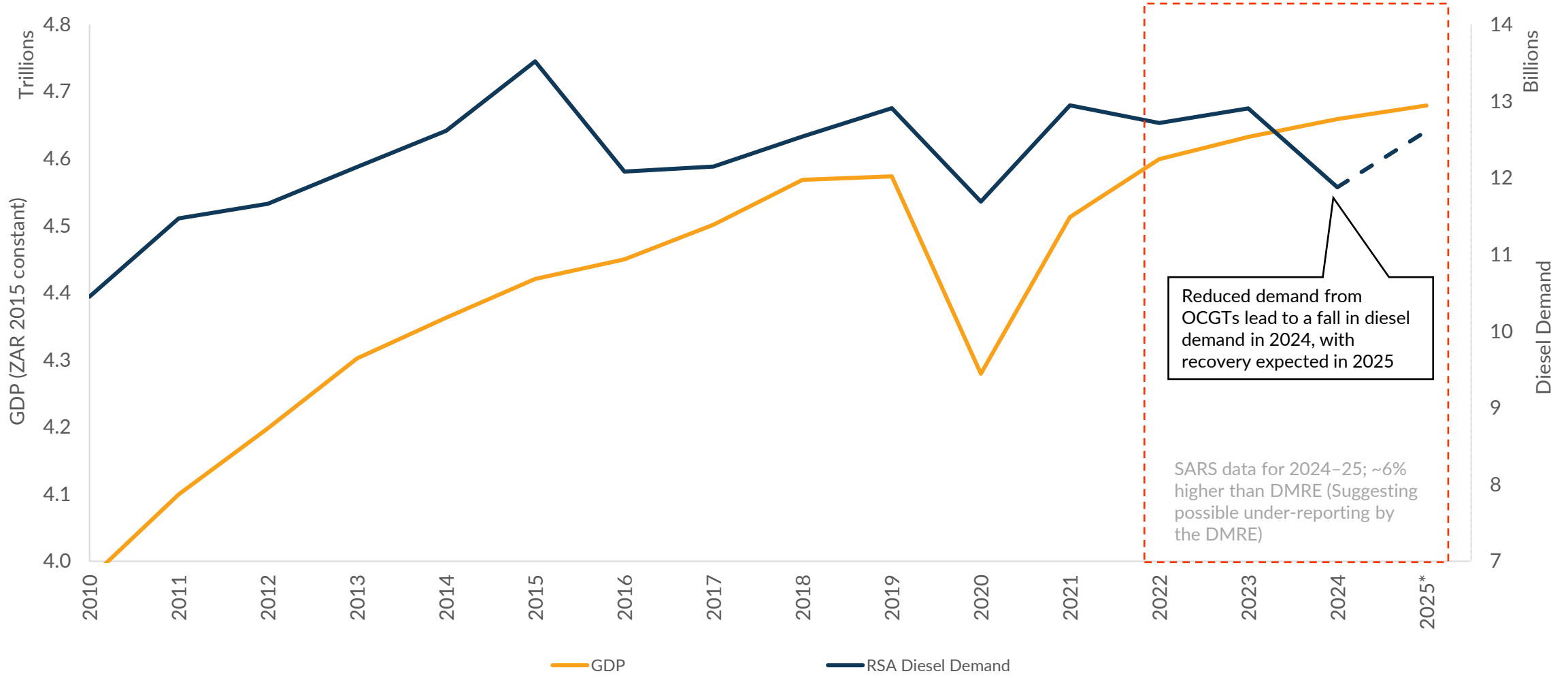
GDP | Power Generation | Rail vs. Road

GDP vs Diesel demand



Sources:. StatsSA – GDP P04412, 2025 GDP annualized ; SARS tariff data

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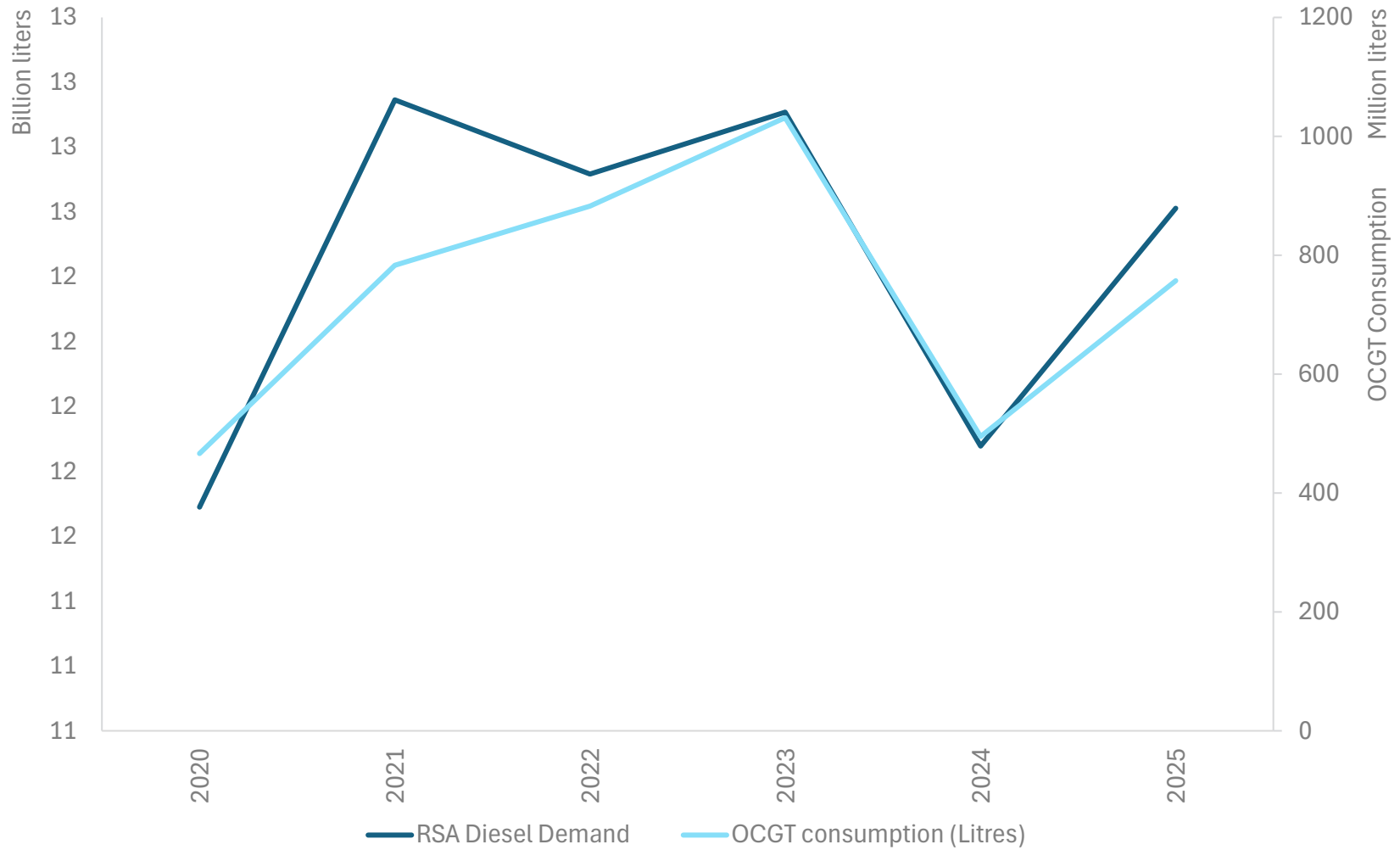
High levels of planned maintenance and delays in restarting Koeberg saw a spike in demand from the diesel peakers….

OCGT Generation and diesel consumption



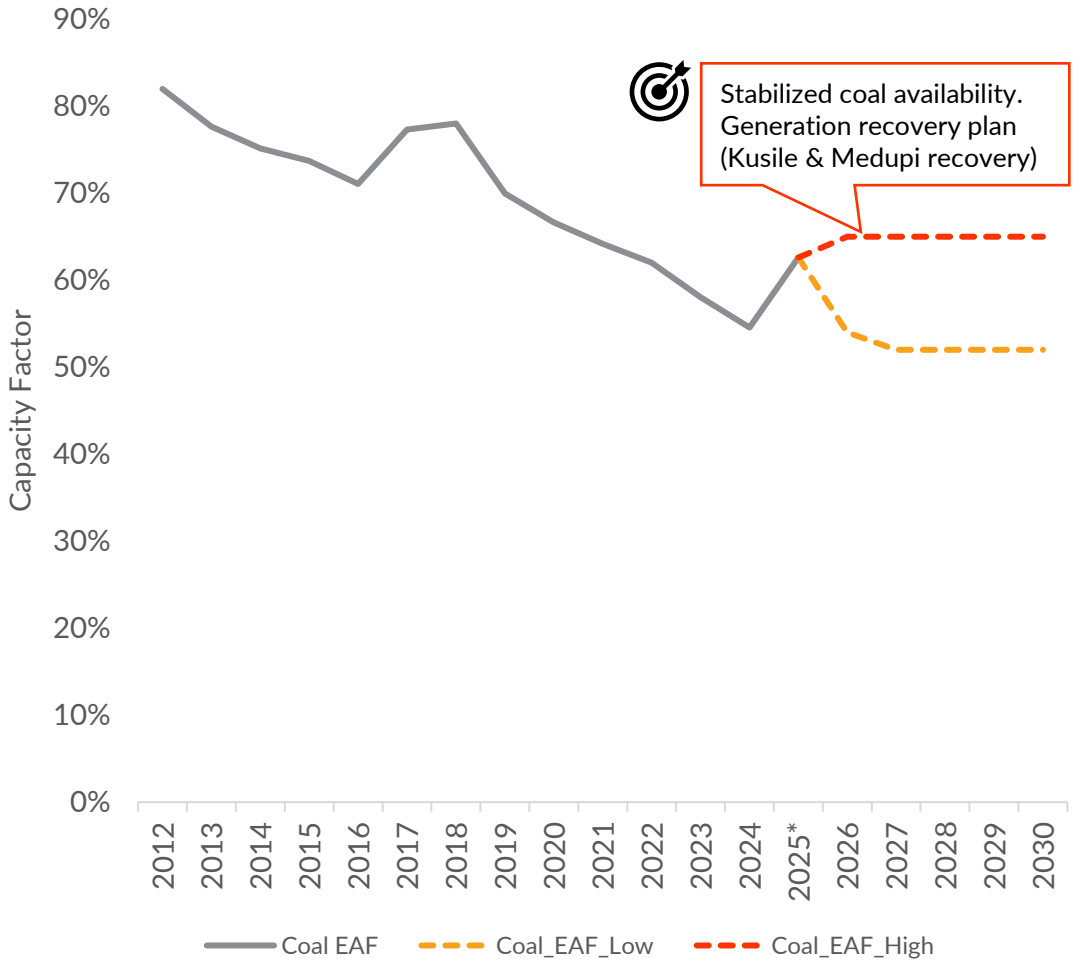
Key points

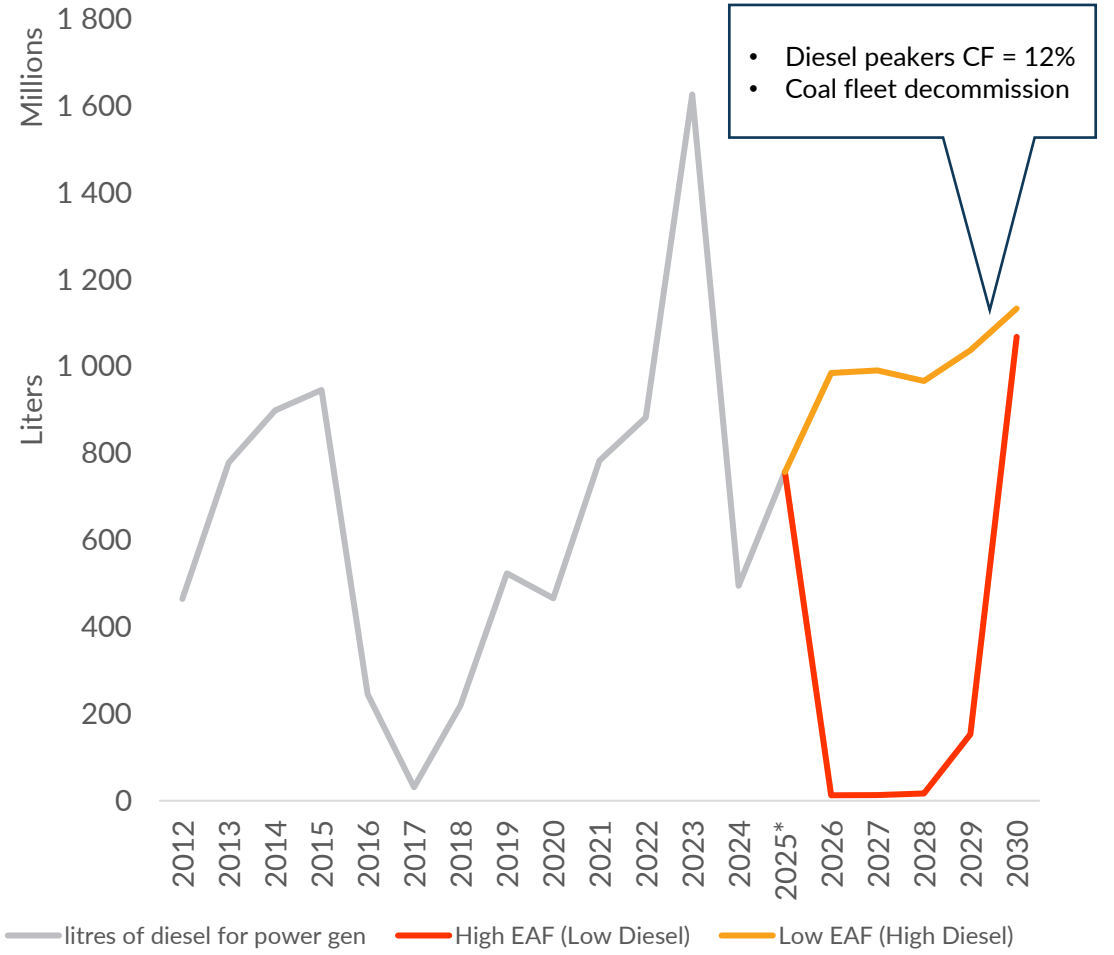
- Eskom + IPP OCGT usage spiked in early 2025 to facilitate planned maintenance to coal generation units as well as favourable diesel prices coming into 2025.
- Eskom uses OCGTs as firm-dispatchable backup for when variable renewable output is low (e.g., in evening or cloudy/low-wind periods). While their load-factor is low they remain a strategic tool to maintain grid stability as renewable capacity expands.
- These measures support Eskom's target of achieving a coal availability factor > **65%** to stabilise long-term supply.

Source: Eskom Data Portal, AIA Analysis, Eskom Integrated Report 2024,FuelsSA

...however we are expecting improved performance from the coal fleet to limit diesel peaker usage in the medium term

Capacity factor of coal power stations at Eskom



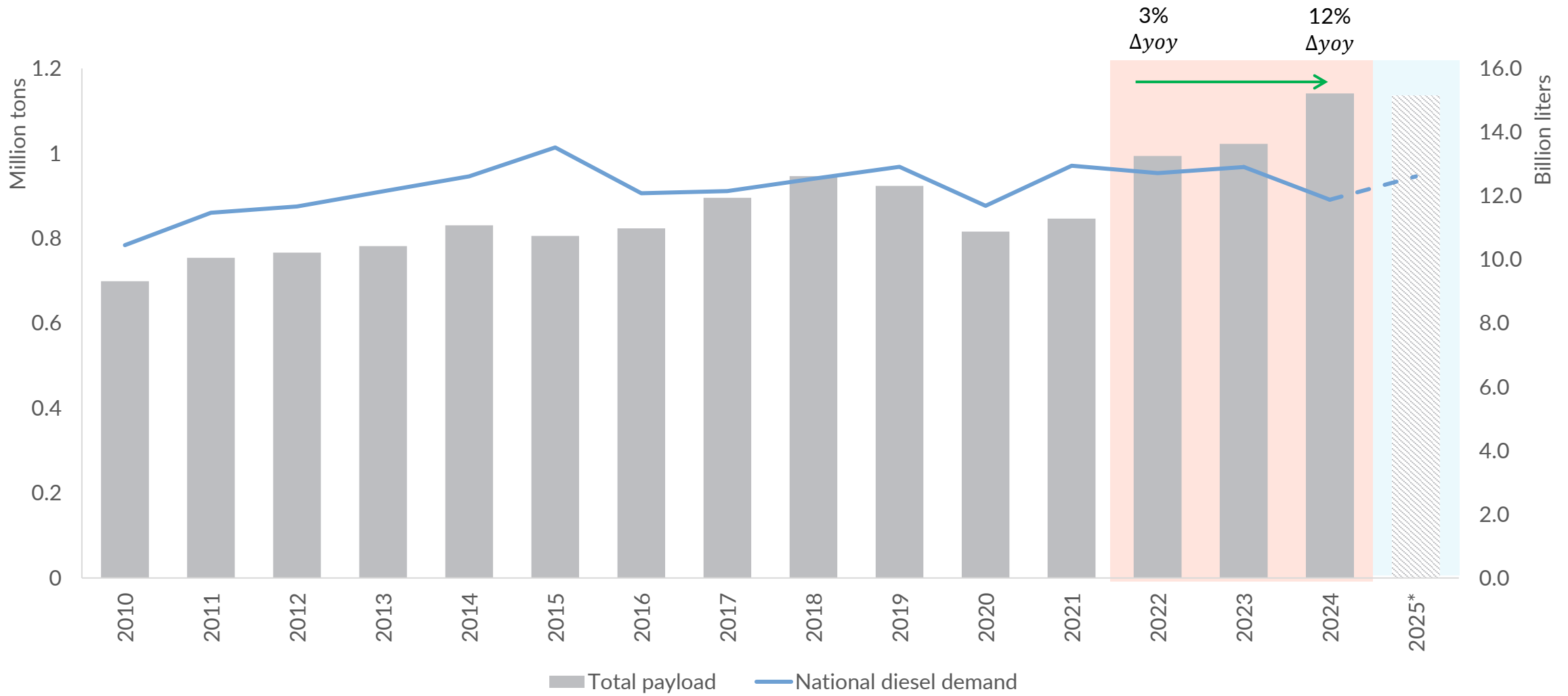


Source: Eskom Annual Reports, Eskom Reports and Generation data; AIA analysis , 2025 annualised

Overall demand for freight rose in 2024 and has since stabilized, however power sector demand has outweighed changes in freight ..

GDP | Power Generation | **Rail vs. Road**

Freight transportation payload

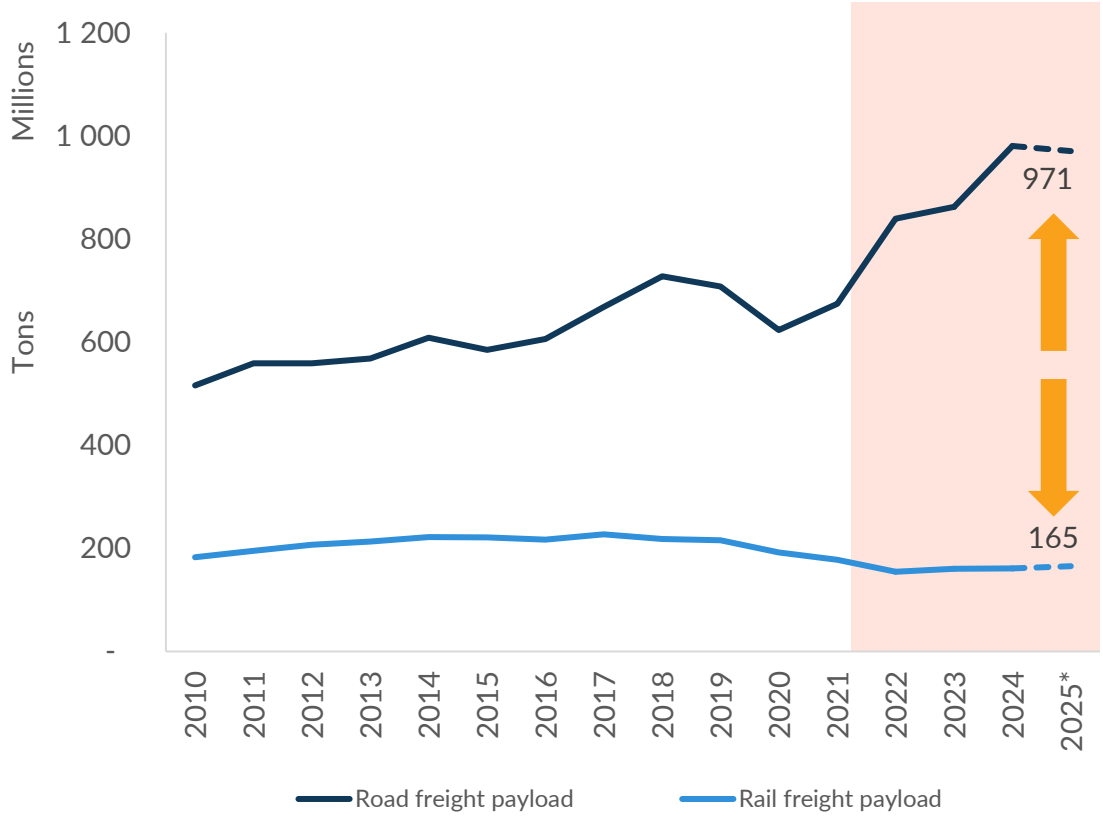


Source: Stats SA Land Transport Survey P7162, Payload & diesel demand YTD 2025 is annualized.

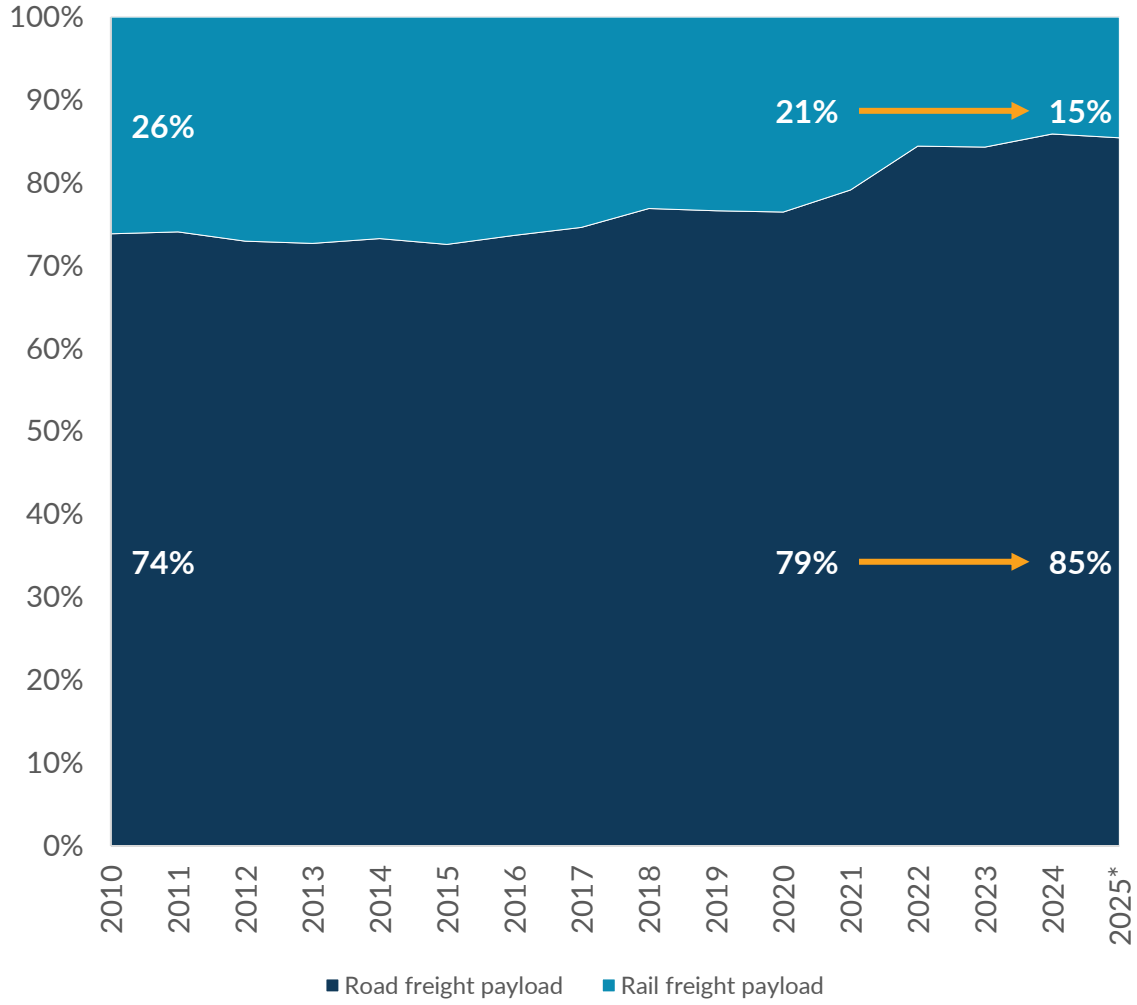
...while we may be seeing the beginnings of Transnet performance improvements impacting road transport in 2025...

GDP | Power Generation | **Rail vs. Road**

Freight transportation payload



Road vs rail freight transportation



Key points

- Transnet aims to restore locomotives through OEM contracts.
- At least 86/394 new generation locomotives restored.
- Rail freight still short of 170 million tons goal due to theft and operational challenges.
- Repairs on container corridors have been completed post-2022 floods

Source: Stats SA Land Transport Survey P7162, 2025 annualised (2025)*

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..with important reforms beginning to be implemented

GDP

Modal Shift Drivers

- Infrastructure & locomotive restoration
- Private operator participation
- Security and Maintenance
- Key freight corridors

Power Generation

Ongoing constraints & recovery

- At least 21% new generation locomotives restored.
- 10 million tons short of 170 million freight by rail tons 2024.

- Long-Term Access Agreements (>10 years)
- 11 Private operators granted access to rail network

- Enforcement
- Monitoring technology
- R14 billion per annum required for maintenance and repairs

- Repairs on container corridors have been completed post-2022 floods
- Container corridor - International Container Terminal Services to continue with Pier 2 Durban upgrades.

Rail vs. Road

Modal shift outlook

Infrastructure	Security	Private operators	Rail corridors
↑	↑	↑	↑

Key Takeaways

- Transnet is rolling out a recovery plan to restore freight rail efficiency. The foundation is set. Infrastructure repairs, access and security reforms, and private sector collaboration are underway.
- Despite progress, freight volumes remain well below targets. Operational challenges and funding gaps continue to limit recovery speed. Reform momentum is positive but not yet restored.
- Looking ahead, Transnet aims to increase freight volumes to 250 million tons by 2029/2030. The path is gradual but structured. Progress depends on sustained investment, accountability, and strong private-sector partnerships.

Rail freight volumes are beginning to recover as Transnet operations stabilize, marking early signs of progress toward network recovery

Sources: Transnet Integrated Report 2025, Reuters, IOL, Transnet Rail Infrastructure Manager: Network Statement

...therefore, diesel demand is anticipated to range between 12-14,4 billion litres, by 2030

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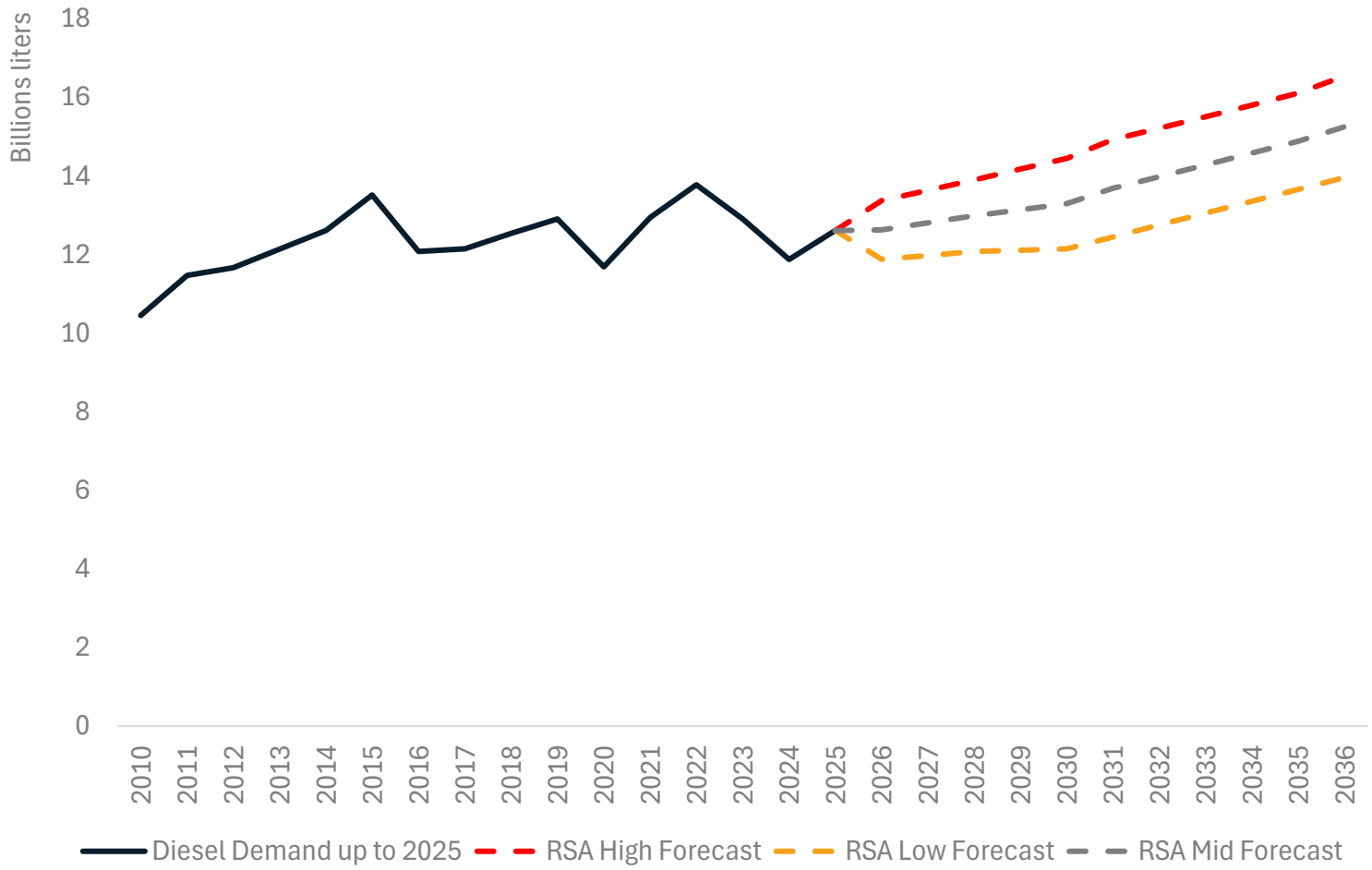
Diesel demand



Assumptions

GDP	Power Generation	Rail vs. Road
↑	↓	↓

- GDP assumptions in line with MTBPS scenarios

Low demand assumptions

- Efficiency increase of 1% pa
- 60m tons of freight moving to rail by 2030
- High EAF as per MTSAO 2024-2028 for OCGT (Avon)
- Electricity supply remains in balance to 2030

High demand assumptions

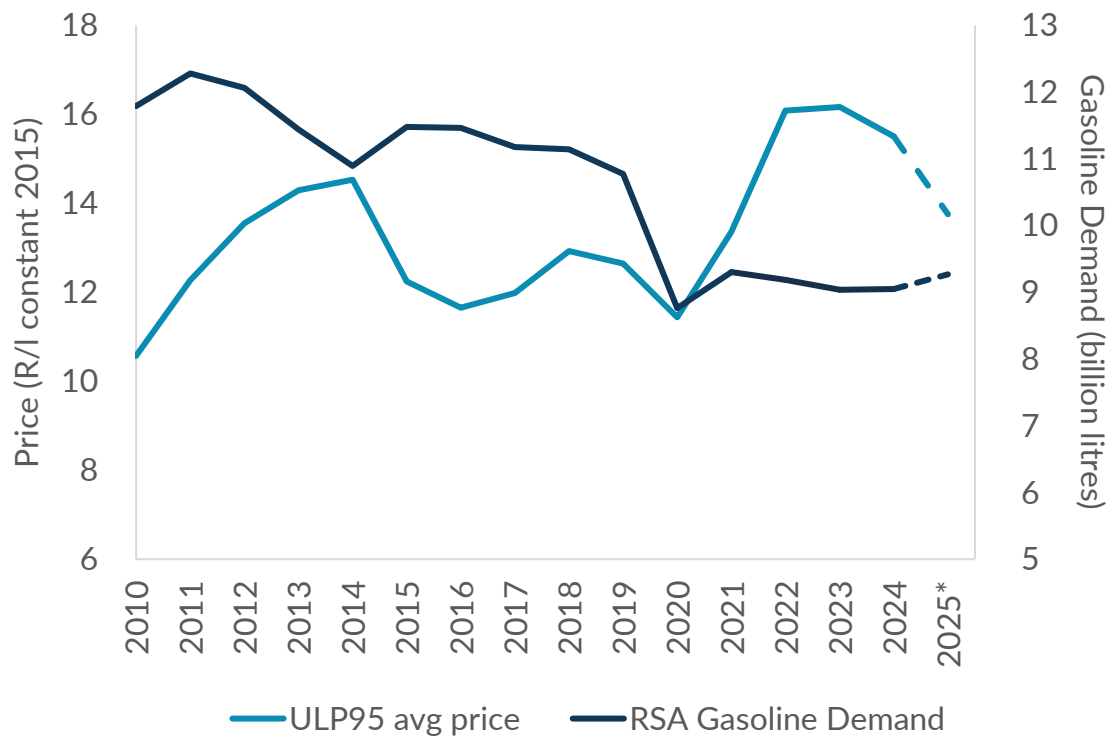
- Efficiency increase of 0.5% pa
- Low EAF as per AIA Analysis for 2026-2030
- Energy gap of 40 743 GWh remains in 2030

Source: 2025 data annualised. Sources: DMRE - SA Fuel Sales Volume data, SARS Tariff Data; AIA analysis

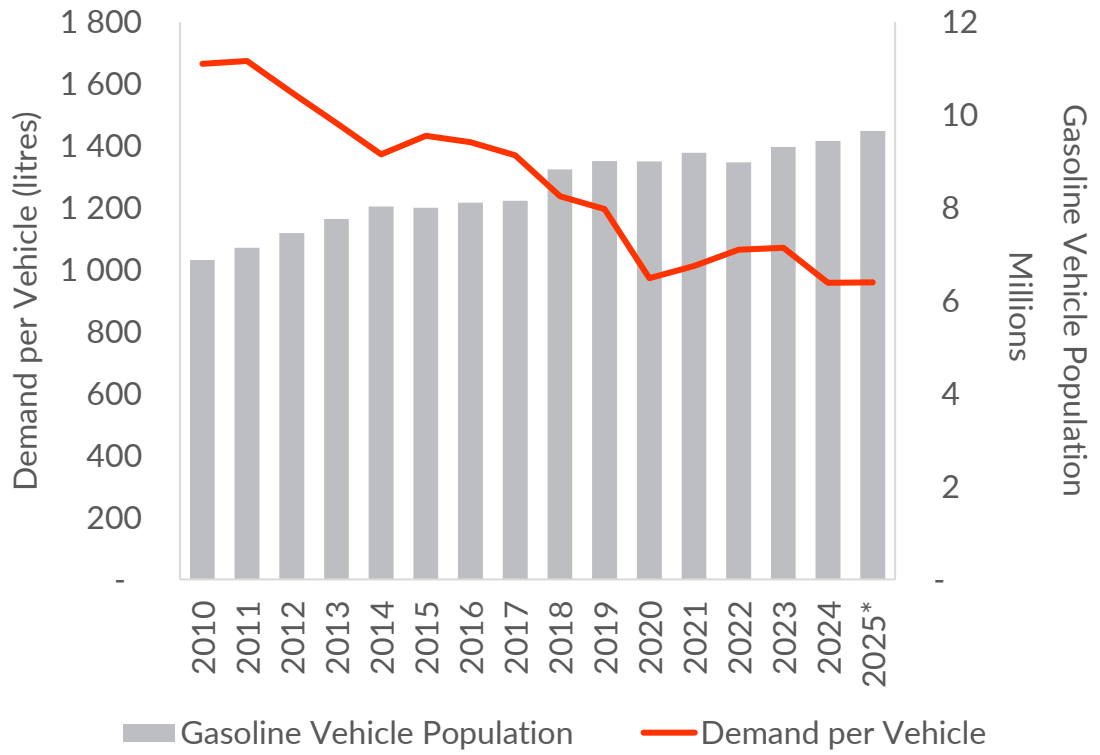
There has been an uptick in national gasoline demand as prices have reduced

Price | Vehicle efficiency | EV Penetration | GDP/Capita and KM travelled | Power Generation

Gasoline price (constant 2015) vs gasoline demand (litres)



Gasoline vehicle population vs demand per vehicle



Key points

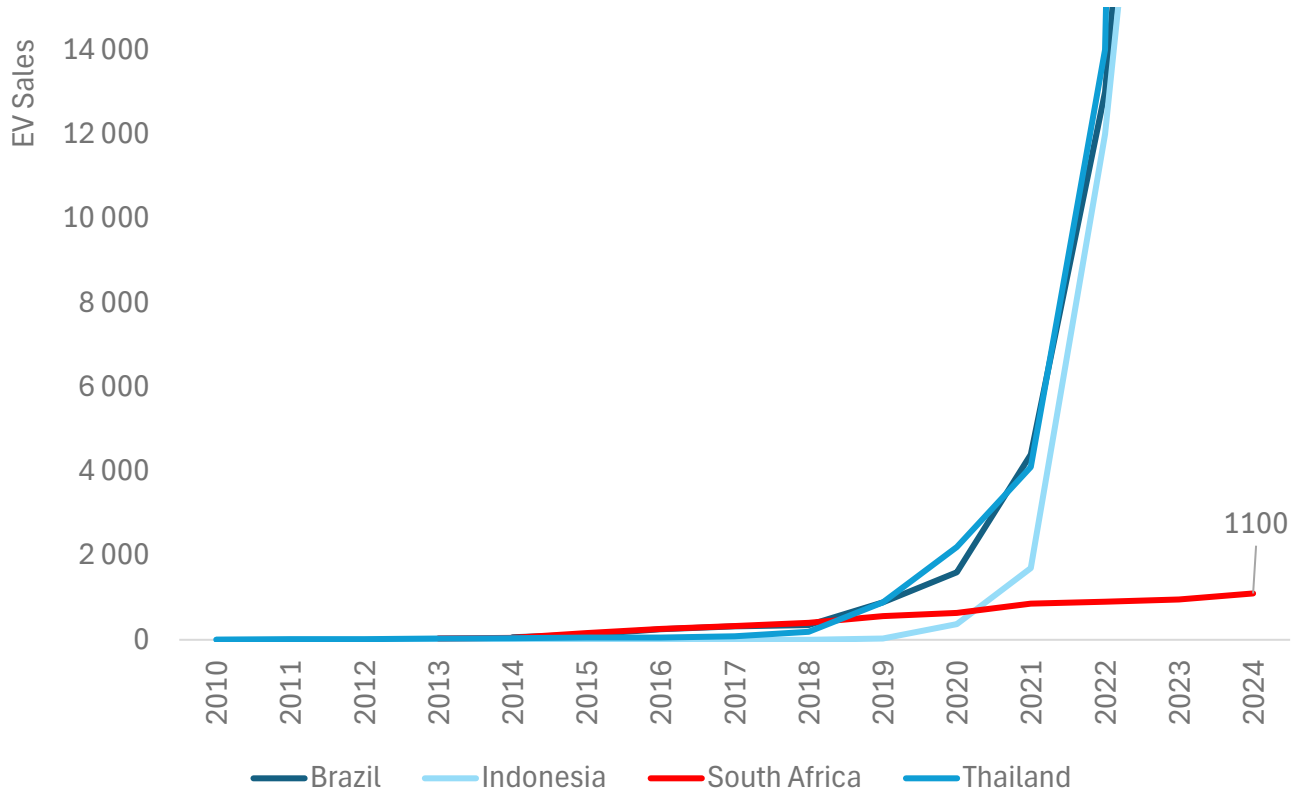
- Gasoline demand rises with lower prices, except in 2019–2021 when COVID-19 restrictions and essential travel shaped consumption.

Sources: DMRE - SA Fuel Sales Volume data, SARS Tariff Data – 2025 is annualized; Gasoline price – AA – Fuel Pricing & Transport Statistics Bulletin; NaTIS – Live Vehicle Population; AIA analysis

South African EV penetration still lags behind comparable economies globally…

Price | Vehicle efficiency | **EV Penetration** | GDP/capita | Kilometres travelled

South Africa – EV sales compared to similar economies



Key Insights

Key Takeaways

- In countries with similar GDP per capita and population, electric vehicle (EV) sales have experienced exponential growth since 2020.
- In contrast, South Africa has lagged, with consumers showing a stronger preference for traditional and plug-in hybrid vehicles. According to Naamsa, these vehicle types have recorded higher year-on-year growth than fully electric vehicles, reflecting limited infrastructure and power supply challenges.
- Once key enablers such as charging infrastructure and reliable electricity supply are established, South Africa is expected to reach a critical tipping point—after which EV adoption will likely accelerate exponentially, mirroring global trends.

Source: IEA Global EV Outlook 2025, Naamsa

..however expanding power capacity and infrastructure are accelerating EV adoption, reducing gasoline demand

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Price | Vehicle efficiency | **EV Penetration** | GDP/capita | Kilometres travelled

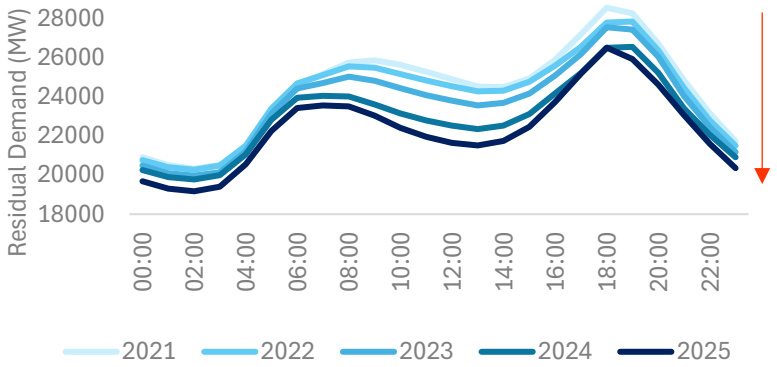
Emerging EV market

Jet fuel demand forecast

A Emerging Power Surplus and the search for New Demand

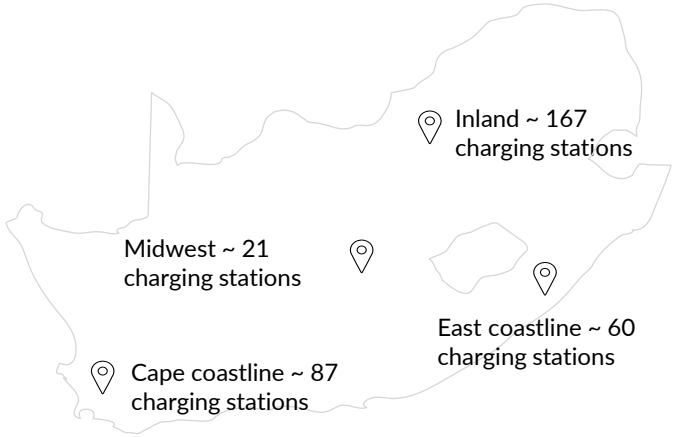
B Geopolitical and infrastructure Investment dynamics

Increasing investment in charging infrastructure



- Surplus electricity, with Eskom & IPP's searching for available markets

Increasing EV investment



- Potential increase in EV uptake due to infrastructure availability
- Chinese BYD to invest 200 – 300 charging stations in 2026 and launching 1 MW fast charging stations

Key points

- The anticipated power surplus positions EV charging as a grid balancing tool
- Strategic measures like tailored time-of-use tariffs and optimised charging station deployment with increased embedded and private solar PV could drive greater EV adoption.
- As charging infrastructure expands, South Africa moves from early adoption to acceleration phase of EV adoption curve
- EV penetration will erode gasoline demand growth over medium term.

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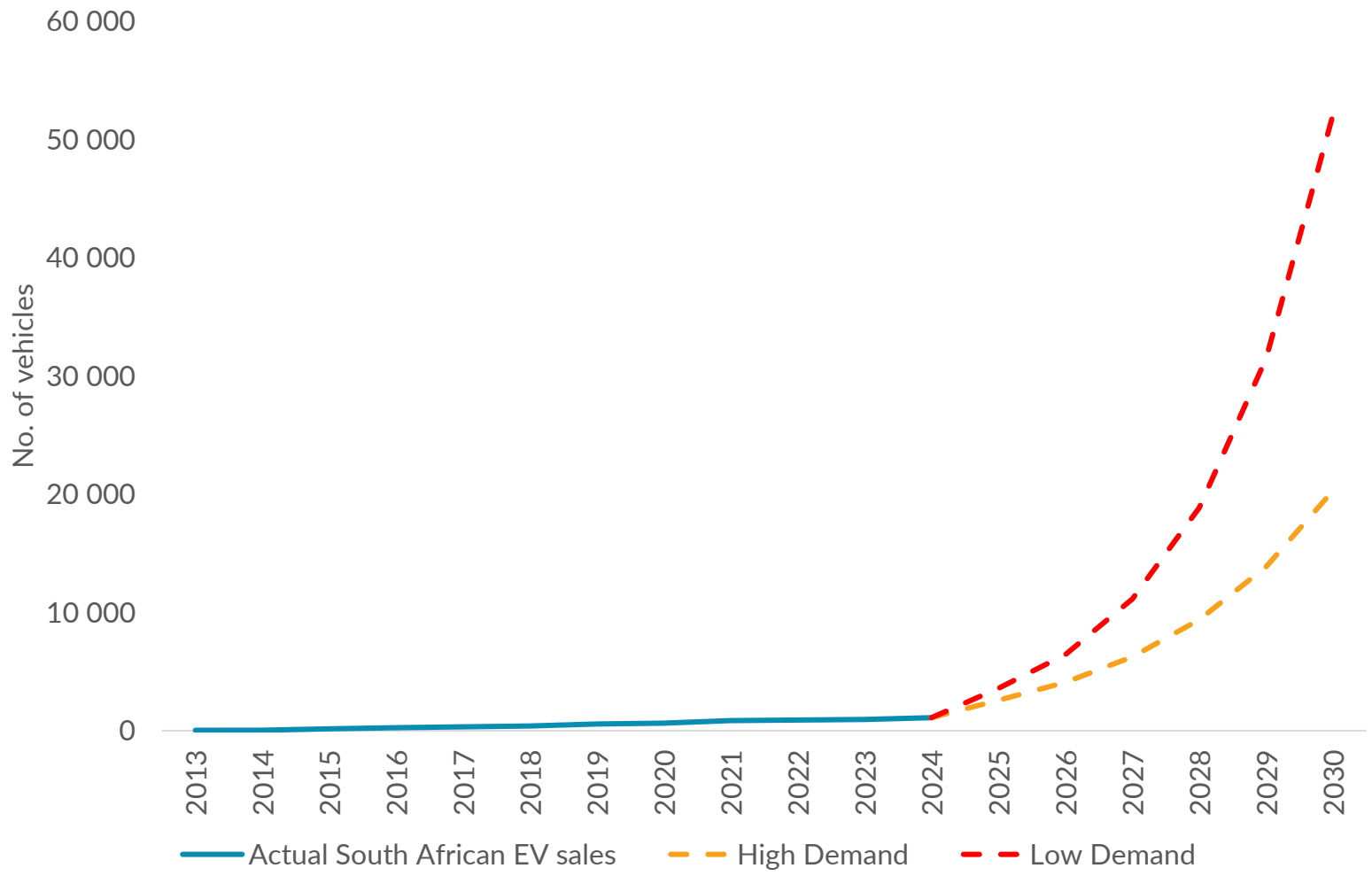
Source: Drive South Africa Interactive Map, IEA Global EV Outlook 2025, Naamsa, Eskom Data Portal Businesstech

...however, the new EV sales in SA remain modest and our assumptions remain unchanged

Price | Vehicle efficiency | **EV Penetration** | GDP/capita | Kilometres travelled

EV Penetration Assumptions



Assumptions

Scenario:	Low	High*
	High EV penetration = Low gasoline demand	Lw EV penetration = High gasoline demand
Rate of Penetration	30% (no change)	90% (no change)
Tipping point (Half-value)	2040 (no change)	2036 (no change)
Slope	3 (no change)	From 3 to 2

*Change is evident due to the base of growth (actual sales have been updated)

...therefore, gasoline demand is anticipated to range between 9-10,2 billion litres, by 2030

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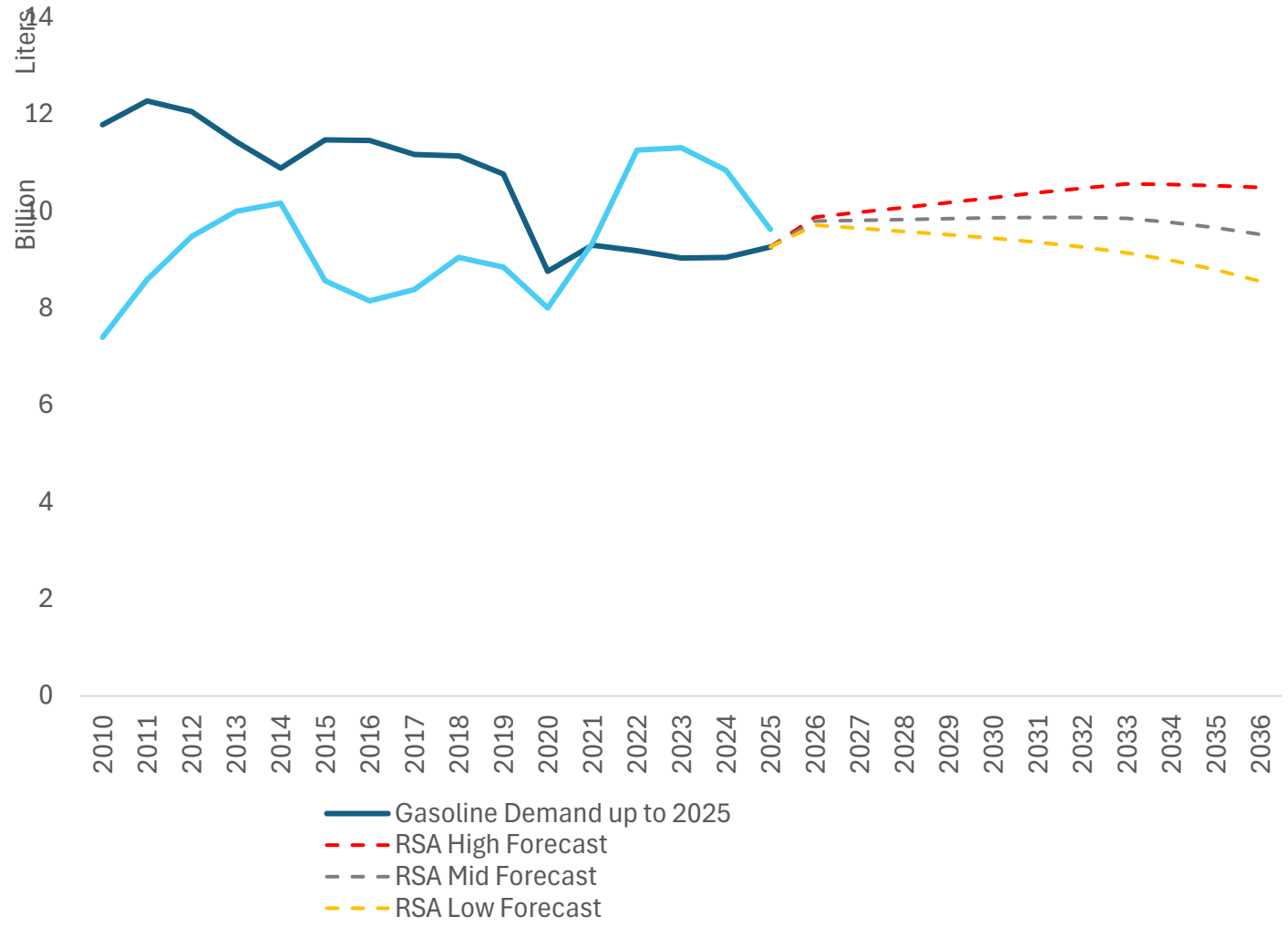
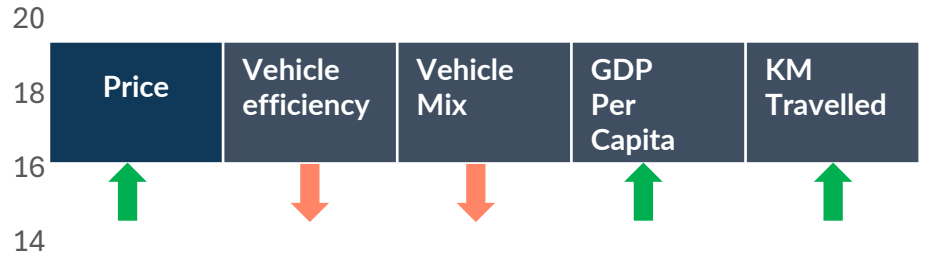
Gasoline demand



Assumptions



- GDP assumptions in line with MTBPS scenarios driving changes in the size of the vehicle parc

Low demand assumptions

- Efficiency increase of 1.5% pa (updated from 1%)
- EVs account for 4% of new vehicle sales by 2030

High demand assumptions

- Efficiency increase of 1% pa (updated from 0.5%)
- EVs account for 1% of new vehicle sales by 2030

Sources: DMRE - SA Fuel Sales Volume data, SARS Tariff Data; AIA analysis

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Overall, global oil surplus production and regional demand drivers are expected to boost overall liquid fuel consumption in South Africa.

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Key Summary

Diesel Demand

Input type	*Insights*	*Outlook*
GDP	South African inflation adjusted GDP showing growth results in growing diesel demand	Economic growth per year supports gradual diesel demand growth.
Vehicle Efficiency	Efficiency gains curb diesel intensity per kilometre.	As efficiencies gradually improve, diesel demand to reduce.
Road to rail	Modest rail freight recovery as Transnet operations improve	Modal shift to potentially trim diesel demand in the medium-term outlook to 2030
Power	OCGTs operating at lower capacity factors as renewable energy peakers.	Diesel demand lower than loadshedding years but potentially sustained by favorable diesel prices.

Gasoline

Input type	*Insights*	*Outlook*
GDP/ Capita	Higher GDP/Capita underpins consumer vehicle purchase and travel	Gasoline demand to increase as affordability improves for citizens.
Price	Falling global oil supply costs and a stronger rand are driving gasoline prices lower.	Gasoline demand to increase as prices come down.
EV penetration & vehicle Efficiency	EVs, traditional hybrids and improving efficiencies curb gasoline consumption	EV infrastructure investment in 2026 has potential to increasingly cannibalize gasoline demand
Increased import	Global oil supply surplus to drive up imports while local supply reduces	Gasoline demand is expected to rise alongside storage demand

Next Steps

- Characterize and quantify vehicle efficiency gains
- Analyze and quantify latest vehicle mix trends (Traditional hybrids, plug-in hybrids, EV's, petrol and diesel)

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Secunda production of liquid fuels is expected to fluctuate in the next 10 years as the gas cliff looms..

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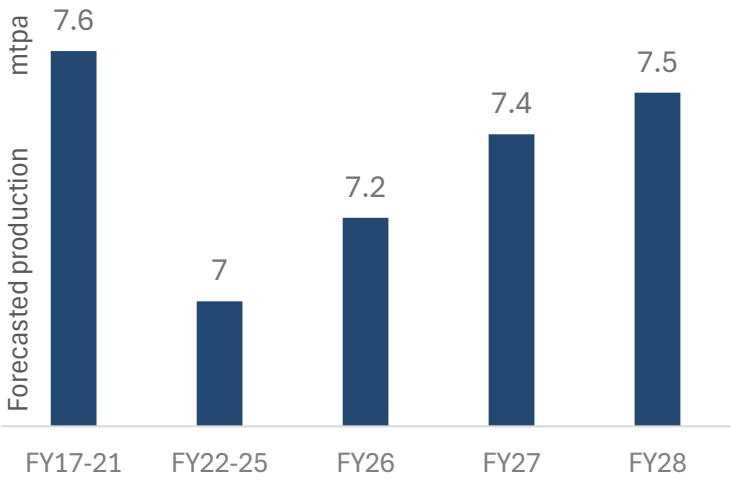
Sasol Coal Quality

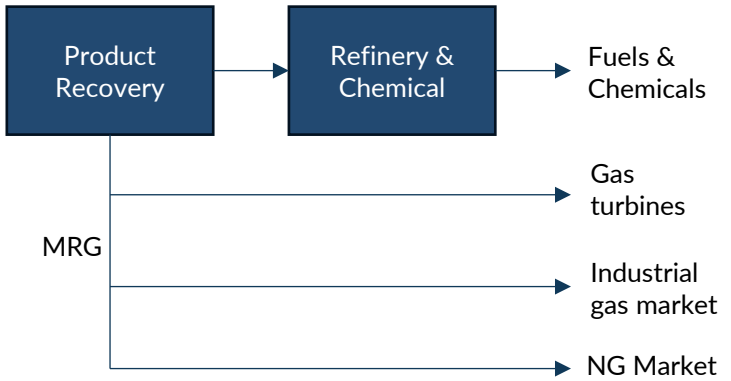
2025 – 2028, indicative



- Destoning plant is expected online by December 2025.
- Secunda gasifier feedstock (coal) has declined in quality in FY22-25 driving the reduction in total production mtpa.
- Gasifier throughput yield and maintenance are directly improved through the removal of non-coal content resulting in **higher production** of fuels and chemicals.

MRG market commitments

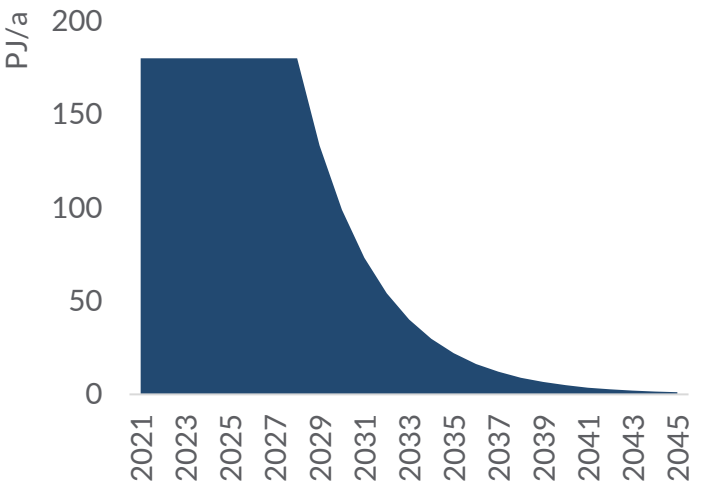
2028-2032



- LNG is unlikely to come online pre-2030 & Sasol has committed to meet the gap in the industrial gas market with MRG produced at Secunda.
- Sasol likely to remove between 21 – 40 PJ /annum of MRG from their gas loop between 2028 – 2030 to supply the market.
- The modifications required to meet this gap may result in **lower production** of fuels and chemicals
- Viability of removing MRG will depend on the gas price currently regulated by NERSA.

Looming Gas Cliff

2032- Onwards



- Depleting gas reserves from Pande Temane to lead to cut-back in liquid fuels production by 2032
- Sustained production between 70% & 80% of production seen as most likely

Source: Sasol

...therefore the forecasted production

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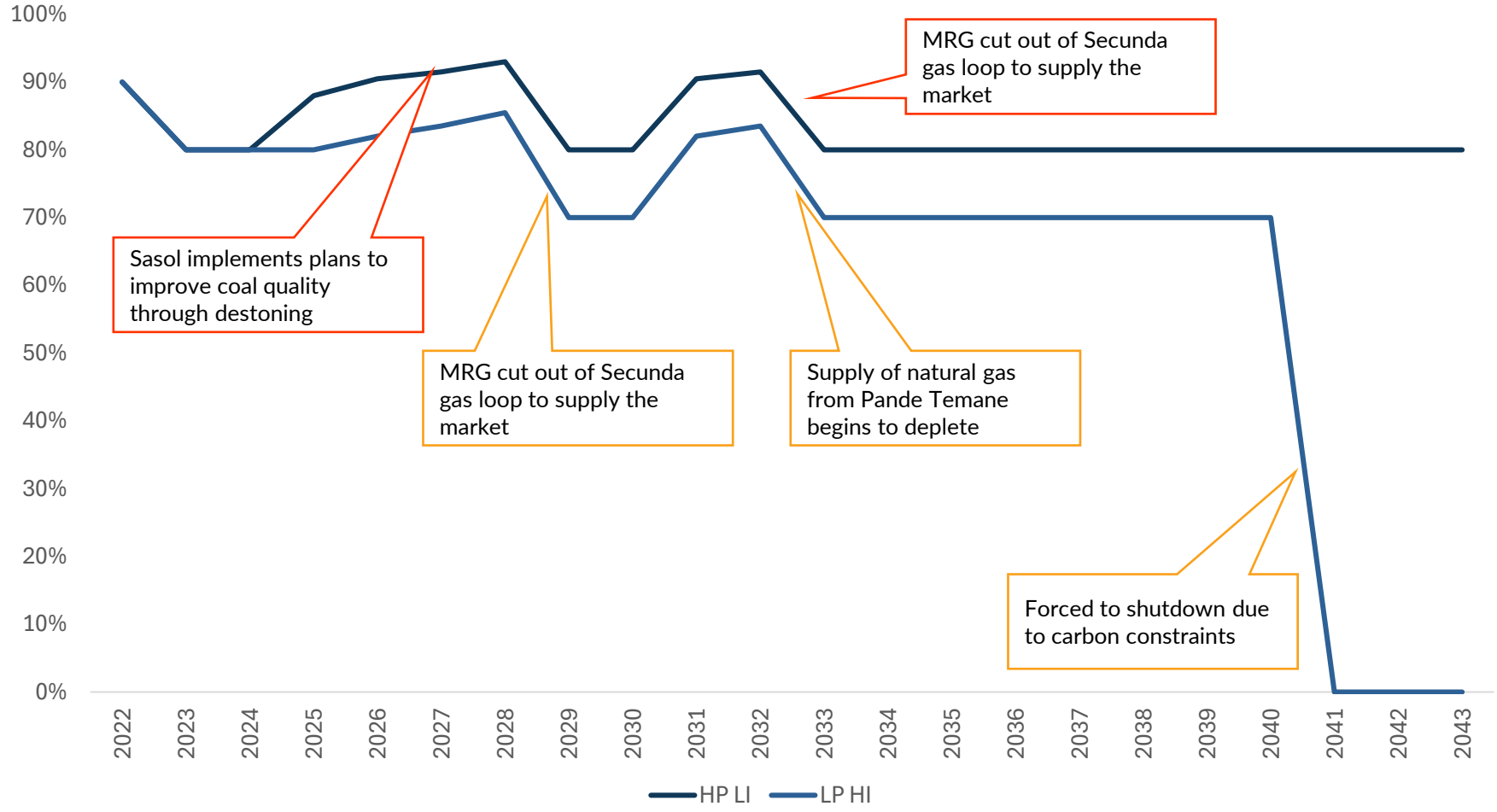
Secunda forecasted production profile (indicative)



Key insights

High production (low import scenario)

- Sasol maintains production due to unlocking yield through destoning.
- Operates at 90% due to improved coal quality and alternative gas supply
- Sasol pulls a maximum 21 PJ/a from the Secunda gas loop.
- LNG coming into the market sets the price 2030

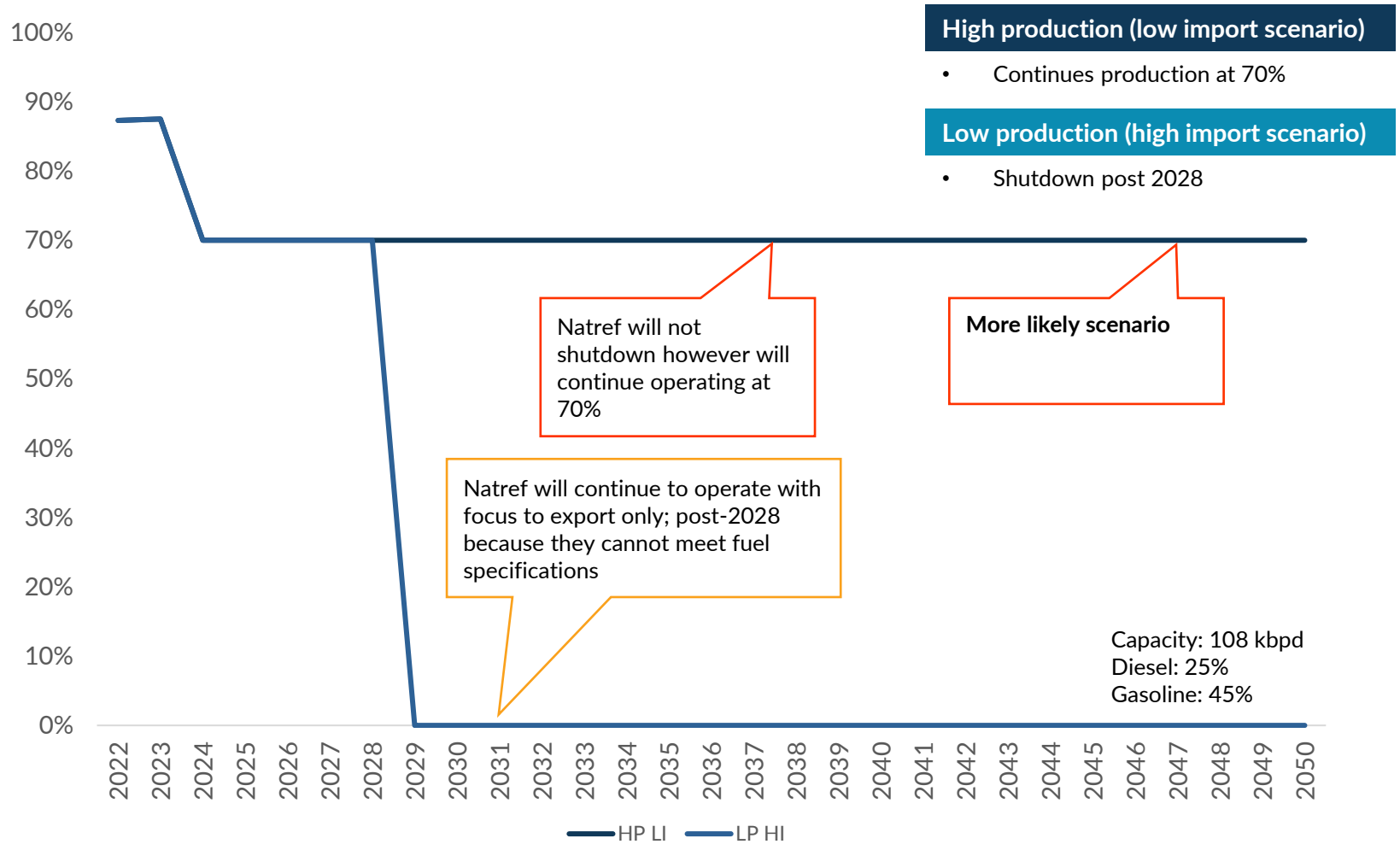
Low production (high import scenario)

- Sasol pulls a maximum 40 PJ/a from the Secunda gas loop.
- Cuts back production to 80% by FY31
- Operates at 70% post 2032
- Shutdown in 2040 due to carbon constraints

...meanwhile, developments towards CF II regulations at Natref have made progress with the installation of a low carbon boiler..

Vopak

Natref forecasted production profile



High production (low import scenario)

- Continues production at 70%

Low production (high import scenario)

- Shutdown post 2028

Key insights

Compliance requirements for Natref:

- Implement a Hybrid Refinery Flow Scheme to meet required CF II specifications
- Upgrade infrastructure to comply with biofuels regulations (BFRF 2020)
- Obtain an amended AEL to include changes from the Air Quality Impact Assessment
- Obtain EA before the AEL application can be approved
- Engineering work is in progress, positioning Sasol to make an FID in 2025

To meet new fuel standards (CF II), Natref is:

- Using low-sulfur crude oil and adding imports (like ethanol) to create cleaner fuels
- Installing a new machine to separate gasoline into cleaner parts
- Adding a special ingredient (FAME) to diesel or processing used cooking oil to make diesel cleaner
- Receiving a special additive (TAME) to blend into fuels, either on its own or mixed with other ingredients
- 2025: Installation of first low carbon boiler and commission of second boiler by end of 2025.

Source: Biofuels Regulatory Framework (BFRF); Natref report: Hybrid refinery flow scheme project – March 2024; Atmospheric Emission License (AEL)

Our view is that the SAPREF refinery is unlikely to restart, however there are ambitions by CEF to build a 400-600kbbl/d refinery

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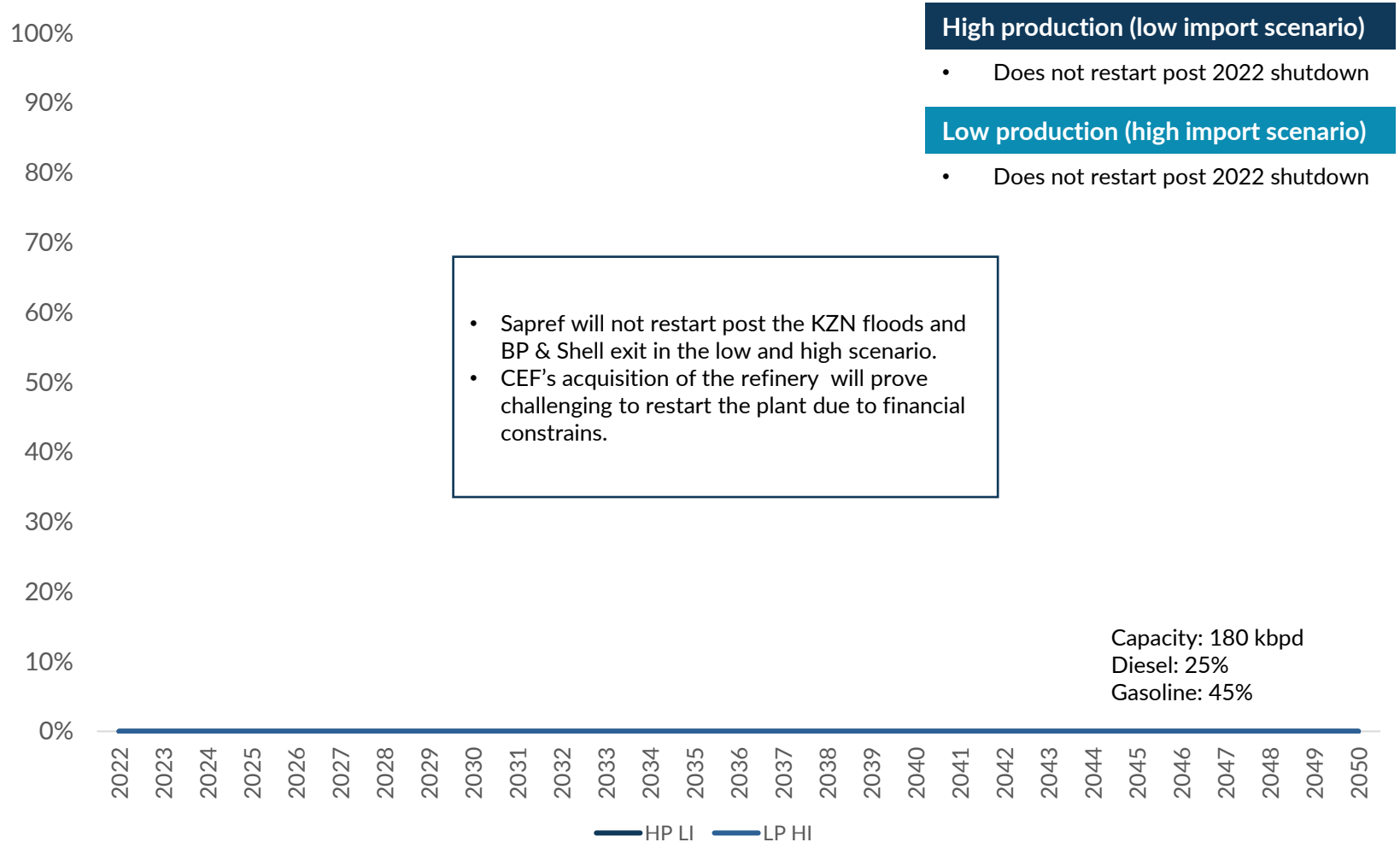
Sapref forecasted production profile



High production (low import scenario)

- Does not restart post 2022 shutdown

Low production (high import scenario)

- Does not restart post 2022 shutdown

- Sapref will not restart post the KZN floods and BP & Shell exit in the low and high scenario.
- CEF's acquisition of the refinery will prove challenging to restart the plant due to financial constrains.

Capacity: 180 kbpd
Diesel: 25%
Gasoline: 45%

Key insights

- Sapref plant shut down after extensive damage to due to the devastating floods in 2022 - refinery submerged in up to 3m of water
- State-owned CEF struck a deal in May to acquire Sapref from BP and Shell for R1
- The government aims to upgrade the refinery to ensure that SA has security of supply
- **The SA government plans to repair and expand the Sapref refinery, increasing its capacity from 400 kb/d to at least 600 kb/d**
- The transaction includes[1] the SDSA and bpSA interests in the Sapref land and other associated assets:
  - Includes: land, crude and finished product tanks, process units, pipelines connecting Sapref to Island View terminal, and the Single Buoy Mooring for crude imports
  - Excludes: bp's marketing businesses, the **Island View terminal operations** and the Blendcor lubricants blending and grease manufacturing plant

CEF aims establish itself as a trader, this could be an opportunity for Vopak to collaborate with CEF

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Source: BP Southern Africa; Shell Downstream South Africa (SDSA); Engineering news

Key producers impacting Vopak are likely stable through 2028, while Secunda faces uncertainty from MRG cutbacks as Pande Temane reserves deplete.

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Supply landscape

Natref (108.5 kbpd)
- Production delays January 2025 – March 2025 following Crude Distillation Unit fire.
- C upgrades underway with commissioned low carbon boilers 2025.

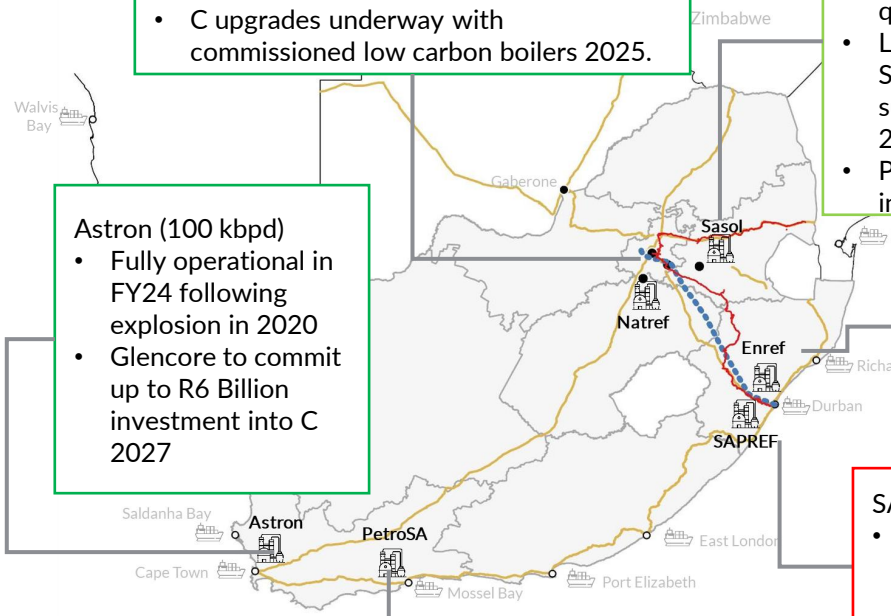
Secunda
- Production to rise 2025–2028 with improved coal feedstock quality.
- Liquid fuels at risk as Sasol maintains MRG supply through 2028–2030.
- Post-2032 gas cliff introduces uncertainty.

Astron (100 kbpd)
- Fully operational in FY24 following explosion in 2020
- Glencore to commit up to R6 Billion investment into C 2027

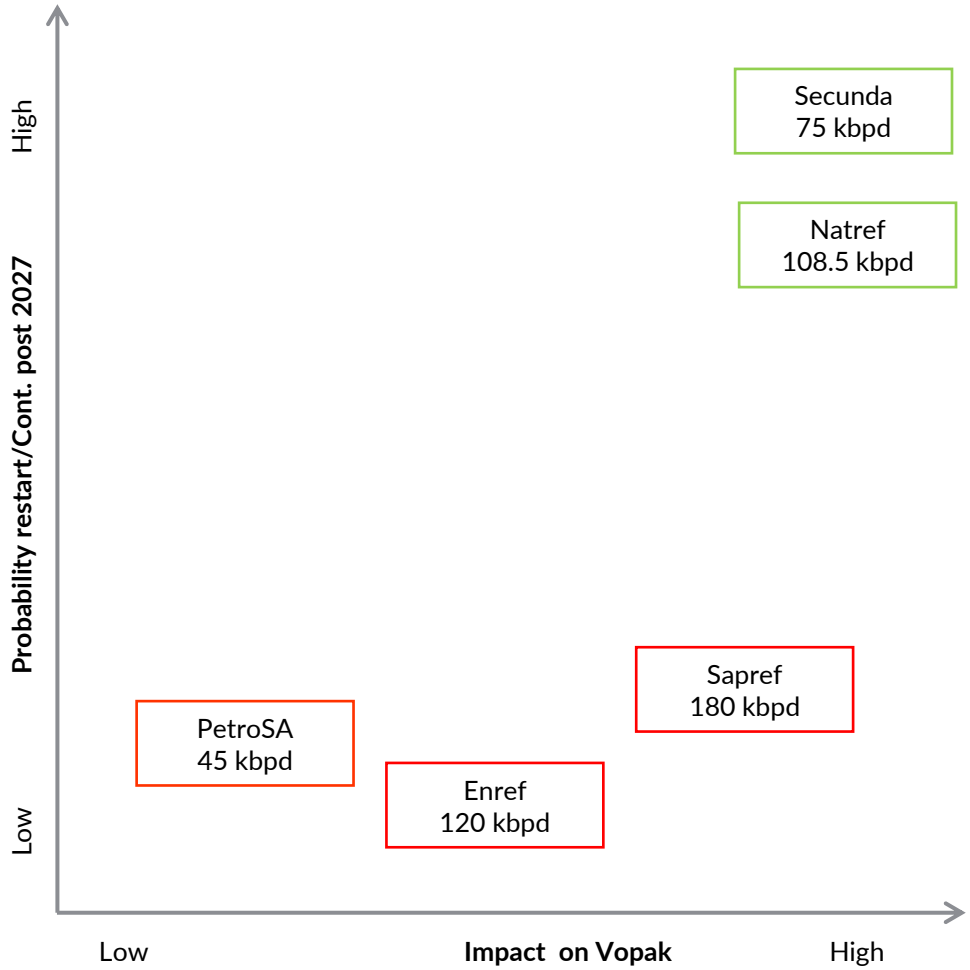
Enref
- Shutdown in 2021 following explosion/fire

SAPREF
- SAPREF acquired by CEF however will not restart due to difficulty with funding

PetroSA
Planned to restart in 2029/30 due to Total exit of 11B/12B fields



Probability-impact overview



Production will continue through 2030, with intermittent output drops and rising import reliance as gas reserves decline.

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Key Summary

Refinery	*Insights*	*Outlook*
Secunda 75 kbpd	In the short term (2025-2028) destoning efforts expected to improve production. Drop in production expected with MRG commitments post 2028. Uncertainty post gas cliff 2032.	Mixed outlook with intermittent rise and drop in production based on gas cliff transitory phase, reduced production post 2032.
Natref 108,5 kbpd	Advancements made towards meeting Clean Fuel regulations in 2025 in spite of joint owner (Prax group) insolvency.	Production to continue at 70% as C upgrades progress
Sapref -	CEF's acquisition of the refinery will prove challenging to restart the plant due to financial constrains.	Unlikely to restart and contribute towards local production

Next Steps

- Develop a high-level carbon balance model for Secunda to assess impacts on liquid fuels production.
- Quantify production changes from 2025–2028 following destoning operations.
- Evaluate the effects of MRG cutbacks and LNG integration on output from 2028–2030.
- Project production changes post-2030, including LNG start-up and the 2032 gas cliff.

AGENDA

- Current market overview
- Demand analysis
  - Diesel
  - Gasoline
- Demand forecast
- Supply dynamics & forecast
- **Namibia-Walvis Bay overview**
- Competitor Analysis – Vitol
- Biofuels Blending Regulations

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Namibia is emerging as the bridge fuelling trade through Walvis Bay and to the copper belt of Southern Africa…

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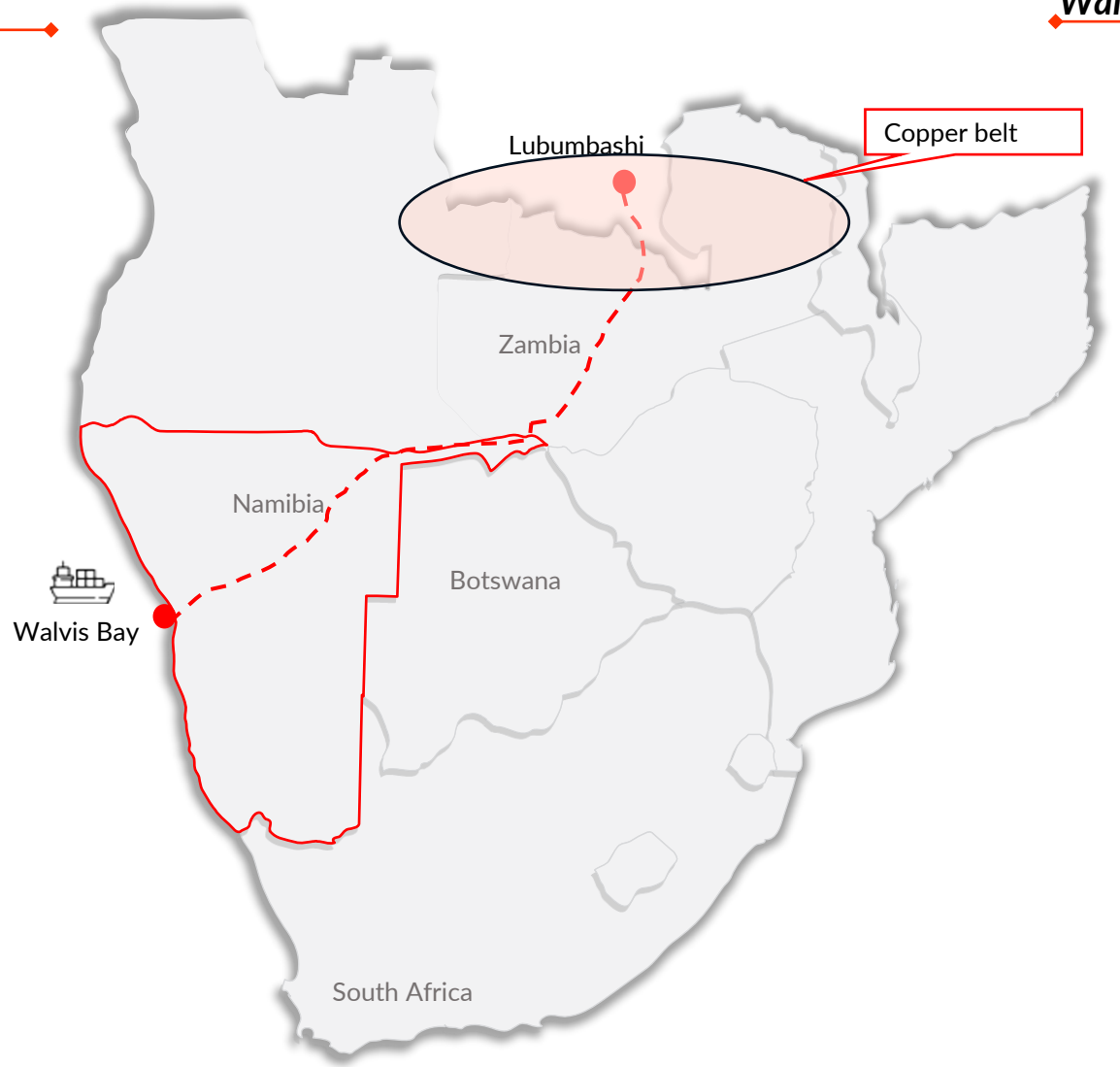
Key drivers of Walvis Bay development

GDP growth & Infrastructure

Oil and Gas opportunities

Renewables

Geopolitics



Walvis Bay as a growing economic node

Greater export volumes into the region and through Walvis Bay

Significant oil and gas production expected online ~2030 flowing through Walvis Bay

Planned gas to power at Walvis Bay to support local generation

Political incentive for alternate trade corridors present Walvis Bay as an opportunity to ensure flow of trade

..with steady economic growth and a growing share of exports in the region..

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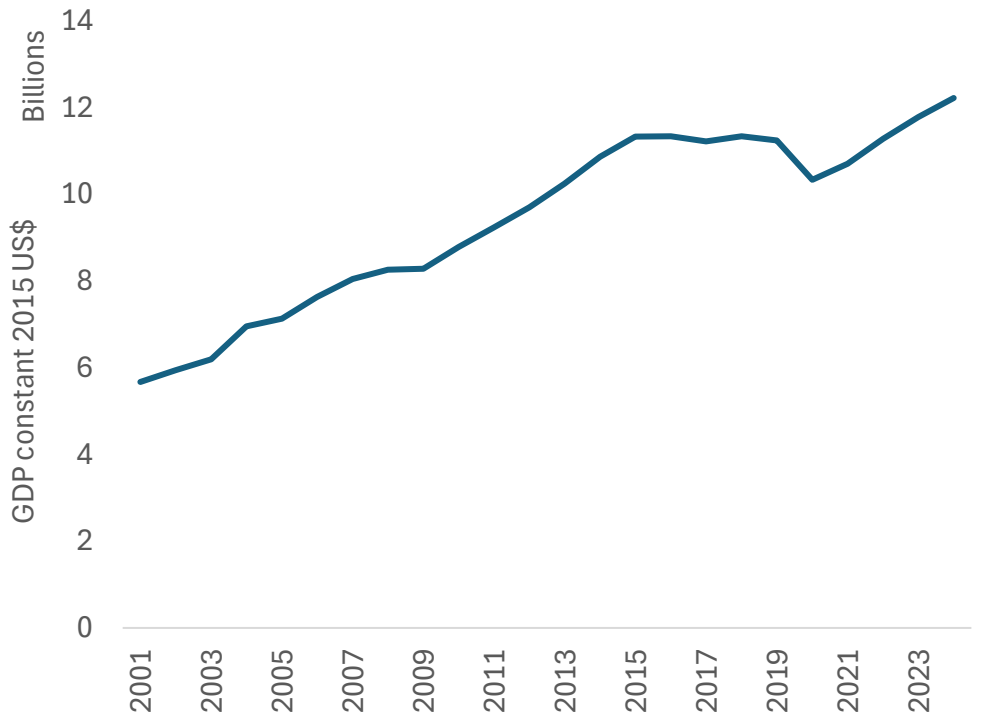
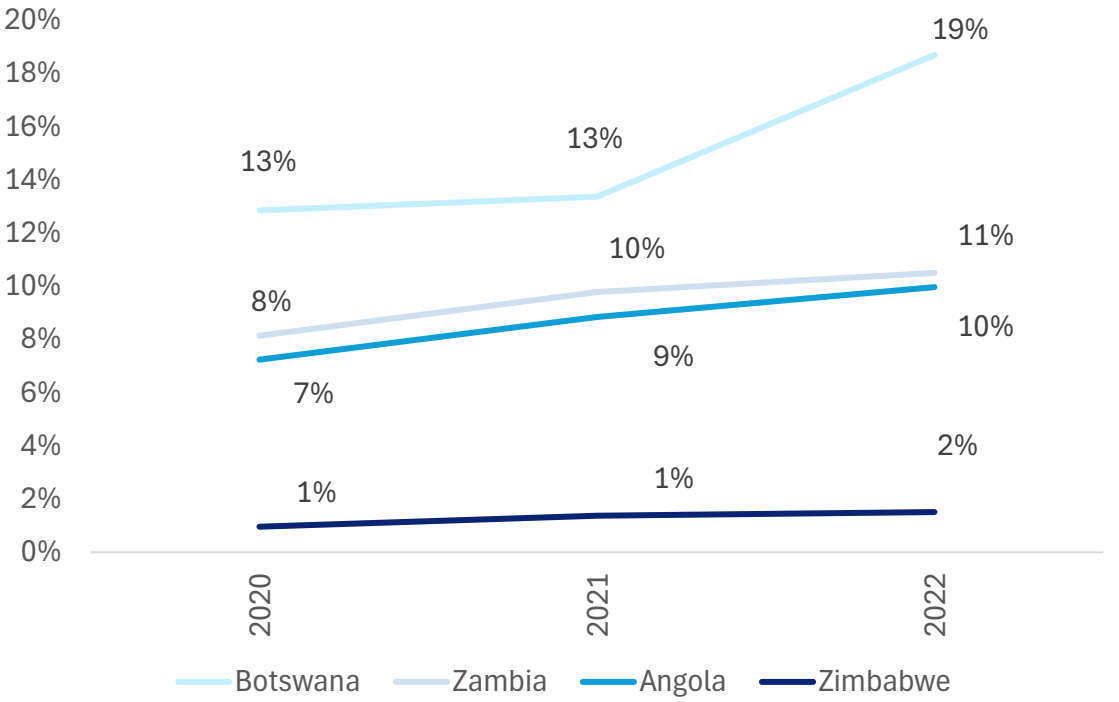
Namibian GDP growth



Namibian % share of exports into Southern Africa



Key points

Namibia's National Trade Policy (2023-2030): Key policy focus areas for growth:

- Diversified export-led growth: Ensure Namibia's exports of goods and services drive growth
- Global competitiveness: Implement strategies to help Namibian firms penetrate international markets.
- Enhanced market access: Create favourable conditions for Namibian firms through bilateral, regional, and multilateral trade agreements
- Integration into regional and global economy: Actively participate in trade initiatives and value chains to support industrialisation and enhanced supply chains.

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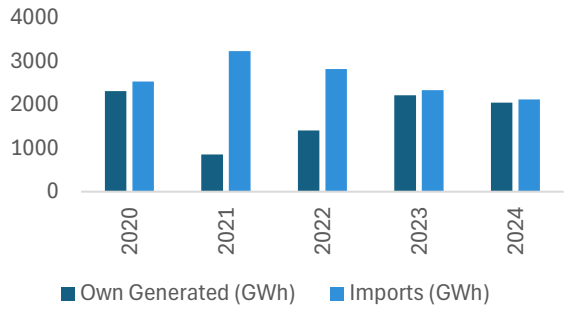
Sources: SARS, World Integration Trade Solution

..as well as a geographic advantage that enables renewables as a viable energy market for local generation and potential power exports.

Namibian is well positioned for renewable energy

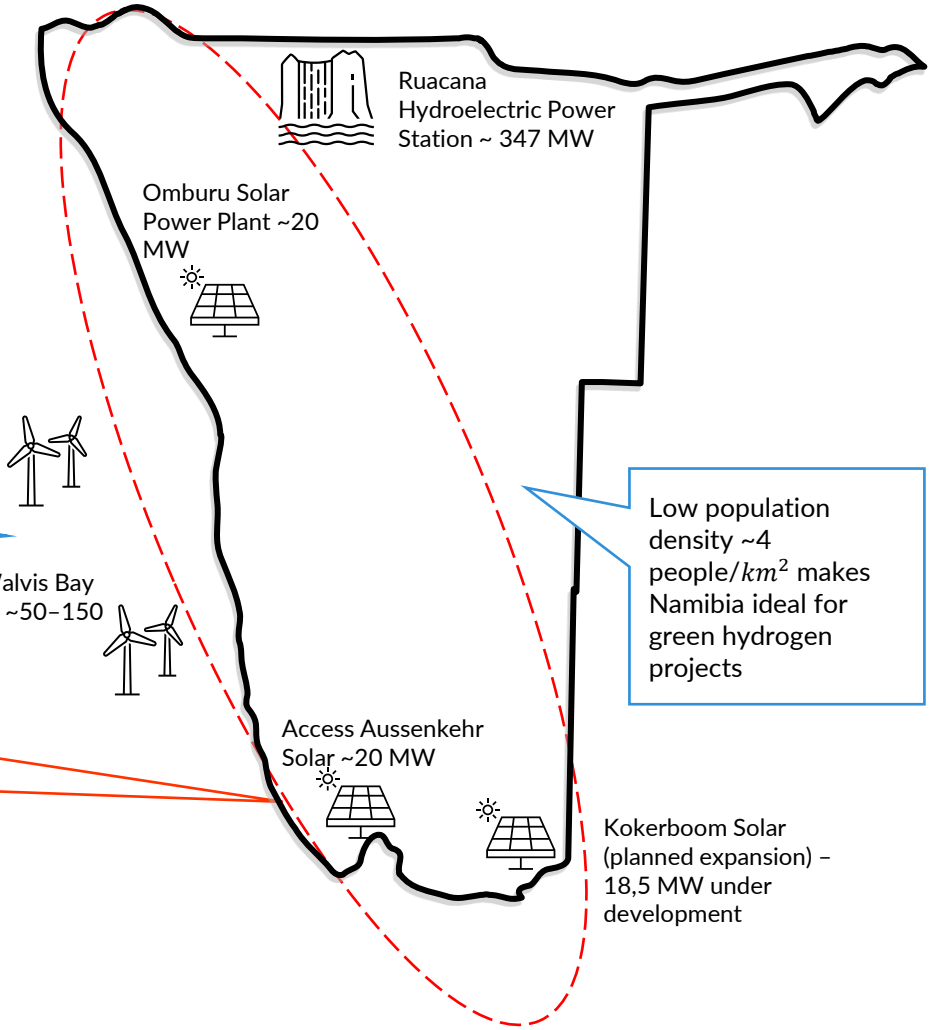
Renewables account for > 90% of Namibian local installed capacity. With a large portion of its electricity coming from imports from South Africa and Zambia.



Strong potential along the southern and coastal belt with average wind speeds > 8 m/s. Feasibility studies underway for large-scale wind farms

Southwestern Namibia has some of the world's highest solar yields, up to 5.8 kWh/kWp per day

Forming an enabling environment for green hydrogen production



Low population density ~4 people/km^2 makes Namibia ideal for green hydrogen projects

Policy Foundation: Vision 2030 positions Namibia as an industrialised nation and drives green industrialisation.

Pilot Projects:
- Hyiron-Oshivela – zero-emission iron exports (Operational)
- Daures Green Hydrogen Village – green ammonia for fertiliser (Pilot completed)

Scale-Up Projects:
- Gigawatt-Scale Hydrogen Plant – hydrogen for green steel, lithium, regional electricity (Pre-feasibility completed)
- Namibia-South Africa Hydrogen Pipeline – cross-border hydrogen infrastructure (Master plan in development)

Export & Infrastructure:
Green Shipping & Ports – enable green ammonia exports (Planning underway)

2030 Goal: Achieve full production of green hydrogen projects, create jobs, and position Namibia as a regional hydrogen hub.

Sources: Namibia Statistics Agency, Namibia's Green Hydrogen Mid-Year Review 2025

Oil and gas reserves off the coast of Walvis Bay position it as a regional liquid fuels and gas gateway..

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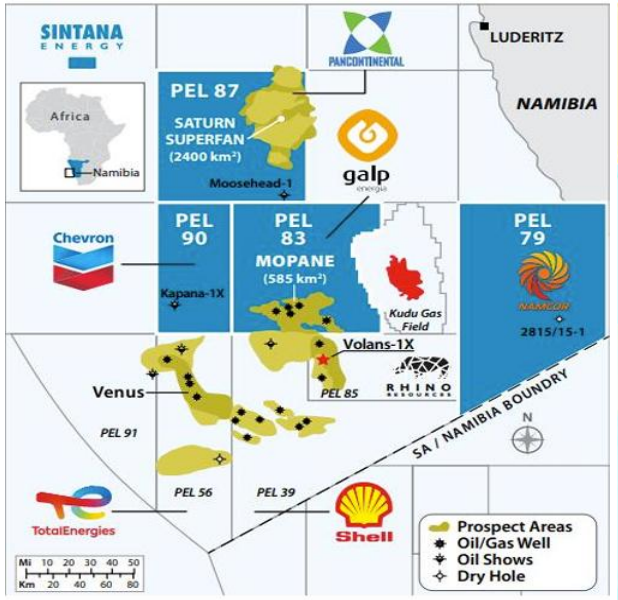
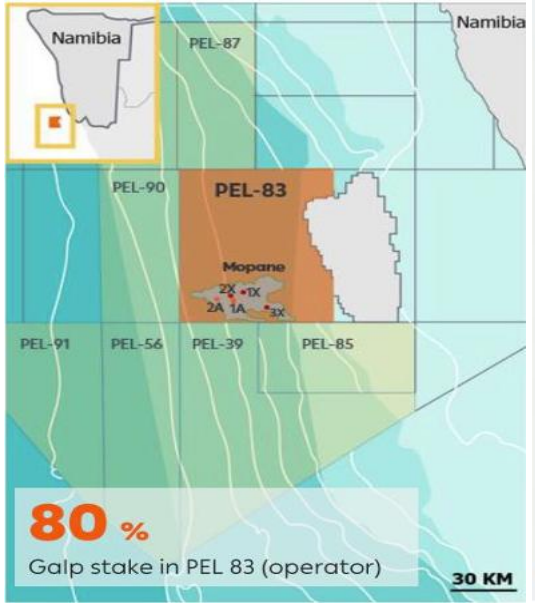
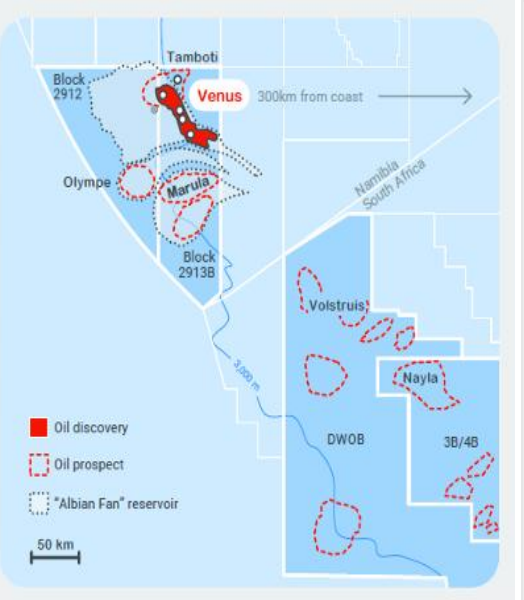
Namibian Block Overview



Galp Pel-83



TotalEnergies



Company	Galp (PEL 83)	Rhino / Azule (PEL 85)	Chevron / Harmattan (PEL 90)	TotalEnergies (Blocks 2912 & 2913B)	Shell	BW Offshore
Wells / Activities	Mopane-1X, 2X, 1A, 2A, 3X	Sagittarius-1X, Capricornus-1X, Volans-1X	Kapana-1X	Venus-1X, 1A, 2A, Marula-1X	Graff-1X, La Rona-1X, Jonker-1A/2A, Lesedi-1X, Enigma-1X	Kharas-1X
Discovery / Volume	Light oil & gas condensates; 10bn BOE in place, 400m recoverable bbl	Hydrocarbons (Sagittarius, Capricornus, Volans); production tests successful	No commercial reserves	Venus: 150,000 BPD; Marula: none	Mixed (some discoveries, some unsuccessful)	Results pending
Timeline	Jan 2024 – Feb 2025	Dec 2024 – Oct 2025	Dec 2024 – Jan 2025	2022 – 2025	2024 – 2025	Sep 2025

Key takeaways

Galp
Multiple Mopane wells confirm light oil and gas condensates (~400 m bbl recoverable). Feasibility studies and 3D seismic support further development. Targeting first oil 2031-2032 with a potential development partner.

TotalEnergies
Deep-water discovery (~150,000 BPD potential) with plans for up to 40 subsea wells and FPSO. FID targeted early 2026; first oil expected 2029-2030, establishing Venus as a cornerstone offshore development.

Rhino/Azule Energy
Multiple discoveries with appraisal wells and flow tests ongoing. FID planned late 2026/early 2027; first oil targeted ~2030.

Key Takeaways:

- Estimated Recoverable Volumes: ~1.5 bn bbl, based on reported recoverable volumes from Mopane (~0.4 bn bbl) and Venus (~1.1 bn bbl), positioning Namibia as a mid-sized, commercially attractive offshore region in Africa.
- Three operators are progressing from discovery to development, with first oil targeted between 2030 and 2032. These projects highlight Namibia's offshore sector as a commercially attractive region with strong potential to drive long-term economic growth.

..which is planned to distribute through the rail and road networks of Namibia into the copper belt and Southern Africa..

Road

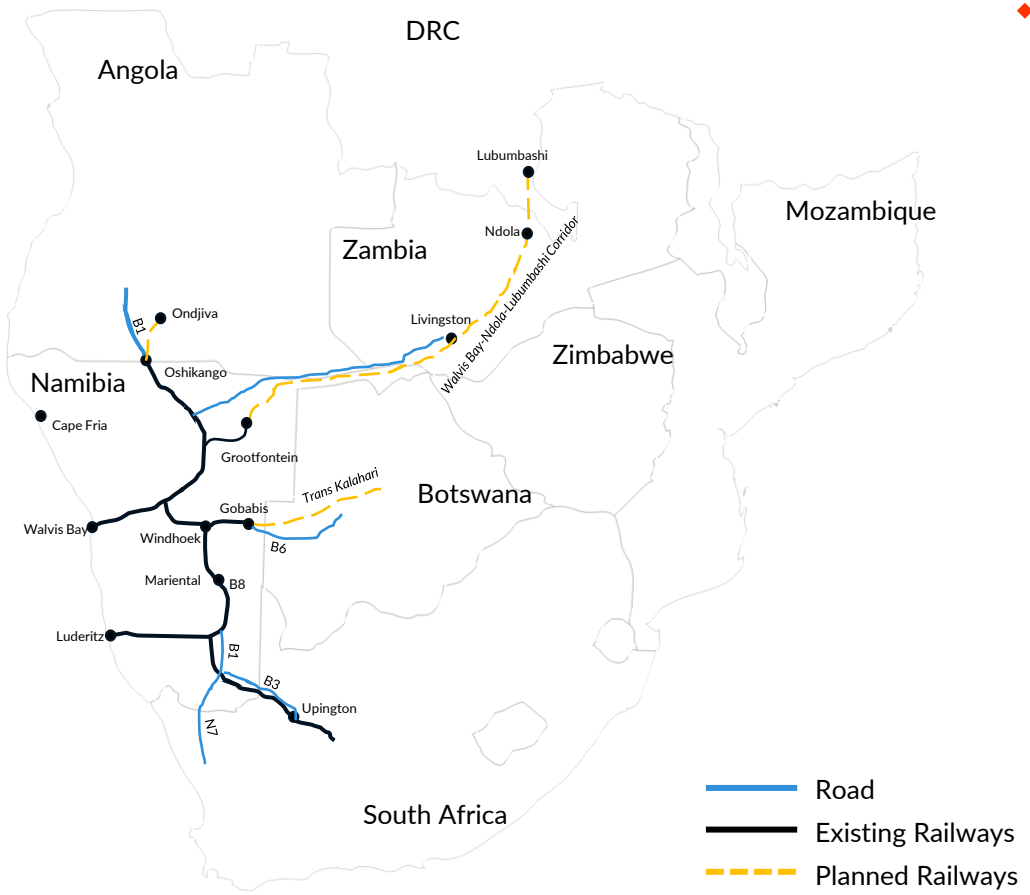
Major road corridors position Namibia as a regional trade gateway, linking the port of Walvis Bay to landlocked SADC countries.

Existing Roads:

- Trans-Kalahari Corridor: Walvis Bay → Windhoek → Botswana → South Africa
- Walvis Bay–Ndola–Lubumbashi Road: Walvis Bay → Katima Mulilo → Livingstone → Zambia / DRC
- Trans-Cunene Corridor: Walvis Bay → Angola border

Strategic Importance:

Supports current trade flows, port access, and regional integration. Vital for construction logistics, workforce mobility, and interim transport until rail projects are operational



Rail

Namibia's rail network forms the foundation for bulk freight transport and long-term regional integration

Existing Rail:

- South Africa: Namibia ↔ South Africa rail link operational
- Angola: Namibia ↔ Angola partially operational (Oshikango–Ondjiva)

Planned Rail Projects:

- Trans-Kalahari Railway (Namibia–Botswana): ~1,500 km, feasibility; for mineral exports via Walvis Bay
- Grootfontein–Livingstone (Zambia): planned Namibia–Zambia rail link
- Trans-Cunene Railway (Namibia–Angola): 400–450 km, under construction

Strategic Importance:

Planned corridors open north-south and east-west trade flows, linking Namibia to Zambia, Botswana, Angola. Supports Namibia's aim to become a regional logistics hub by 2030.

Namibia is positioned as the strategic corridor from Walvis Bay into the DRC's Copper Belt and onto global trade.

..this supply of liquid and gas fuel can meet several regional offtakers in Southern Africa, including the Northern Cape.

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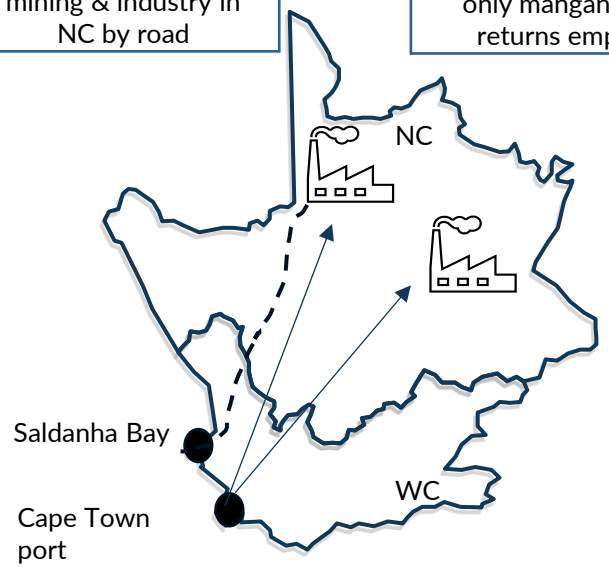
Northern Cape

70% of Cape Town port liquid fuels supply mining & industry in NC by road

Rail freight to Saldanha Bay carries only manganese, returns empty

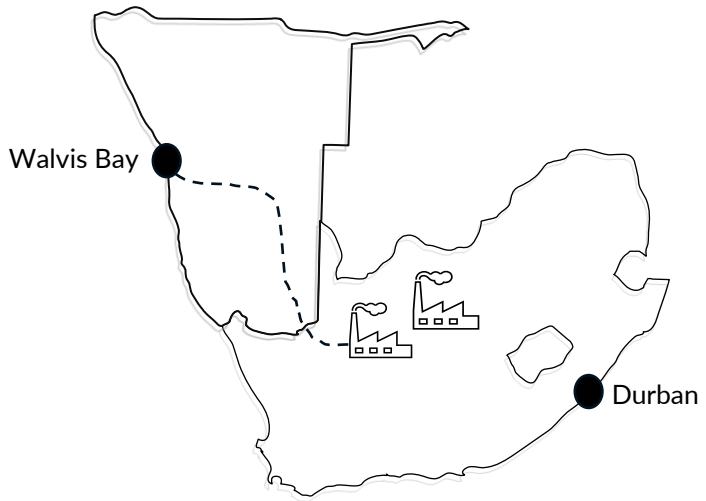


- Cape Town port cannot meet increased demand alone.

Walvis Bay

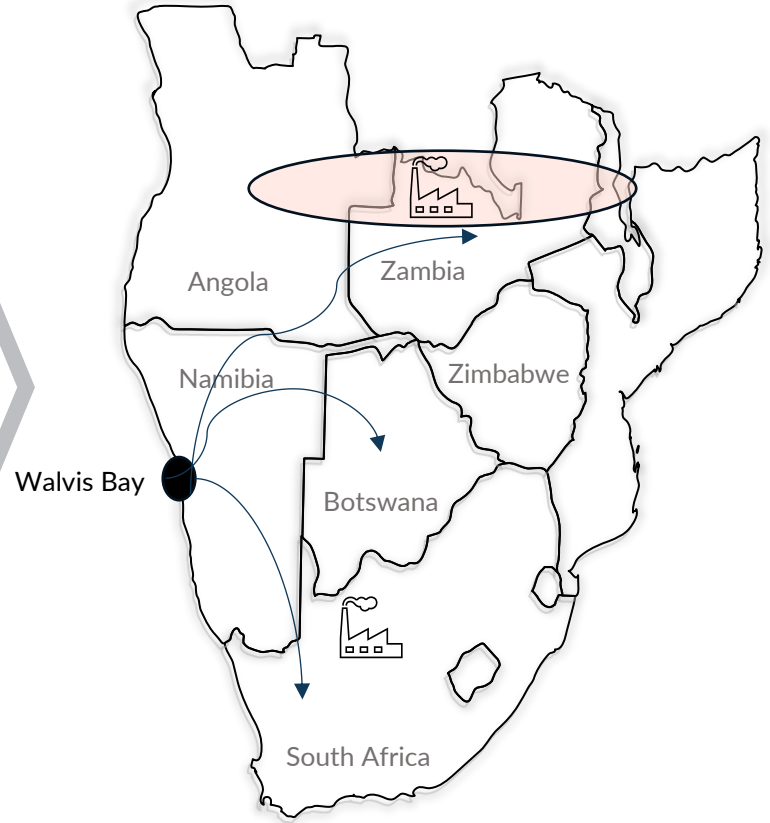
The liquid fuels coming into Walvis port have a potential offtake in the Northern Cape

Alternatively the demand could be met by Durban port from the east



- Walvis Bay liquid fuel supply can meet the demand of a growing mining sector in the Northern Cape.
- Price dependent on the economics of freight from Durban port

Southern Africa and the copper belt



Similar offtakers can be identified in the Southern Africa region and into the copper belt

Additionally these reserves can meet the growing demand for independently generated power within Namibia.

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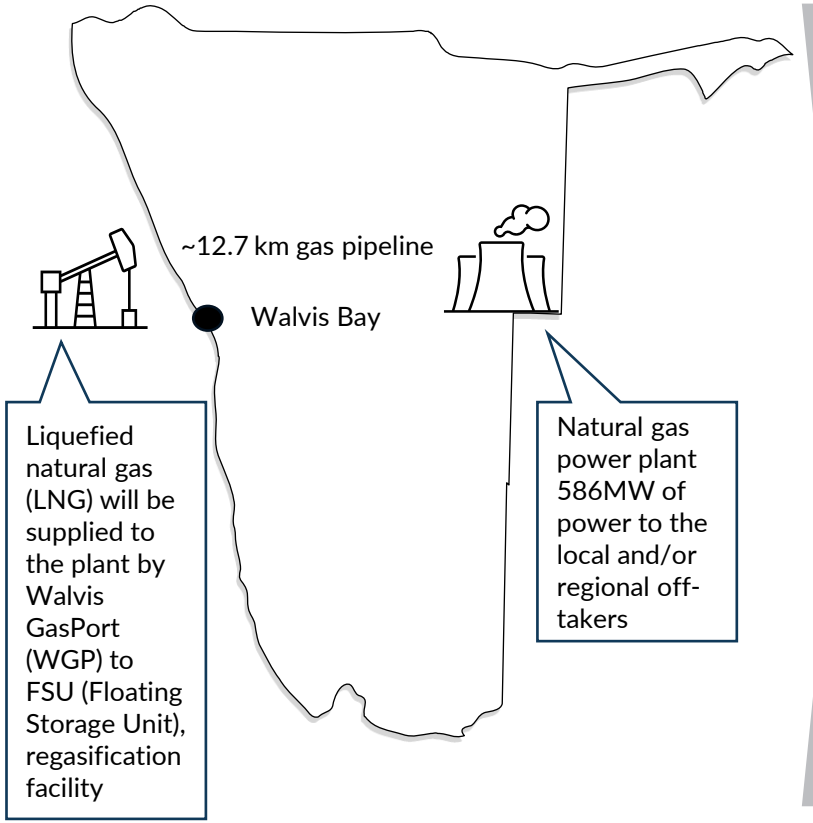
Namibian reliance on imported power

Namibia had a contractual import arrangement of around 180 MW from Zambia.

The reduction is due to Zambia's lower hydro-power generation caused by drought and transmission/interconnector constraints.
Zambia reduced export capacity to Namibia from 180 MW to 120 MW, a 60 MW reduction.

Zambia's cabinet instructed ZESCO to recall at least 100 MW of power from exports to meet domestic demand.



Implications for Namibian generation

Boosting domestic generation: The planned LNG terminal and ~586 MW gas-turbine power plant will reduce Namibia's reliance on electricity imports from neighbouring countries.

Strengthening the corridor: Strengthens the port's energy hub role, making the corridor not just about minerals or goods but also energy infrastructure.

Reducing vulnerability: Increased self-sufficiency lowers exposure to external shocks, such as droughts affecting hydro power or disruptions in import contracts.

Fiscal benefits: Lower import dependence decreases foreign currency outflows, improving the national budget and utility finances.

Enabling industrial growth: Reliable, local power creates conditions for industrial development and export-oriented value chains along the Walvis Bay corridor.

Sources: nmpowerplant, powerinvest.net

Finally, global politics incentivize the creation of alternate corridors such as to move critical minerals.

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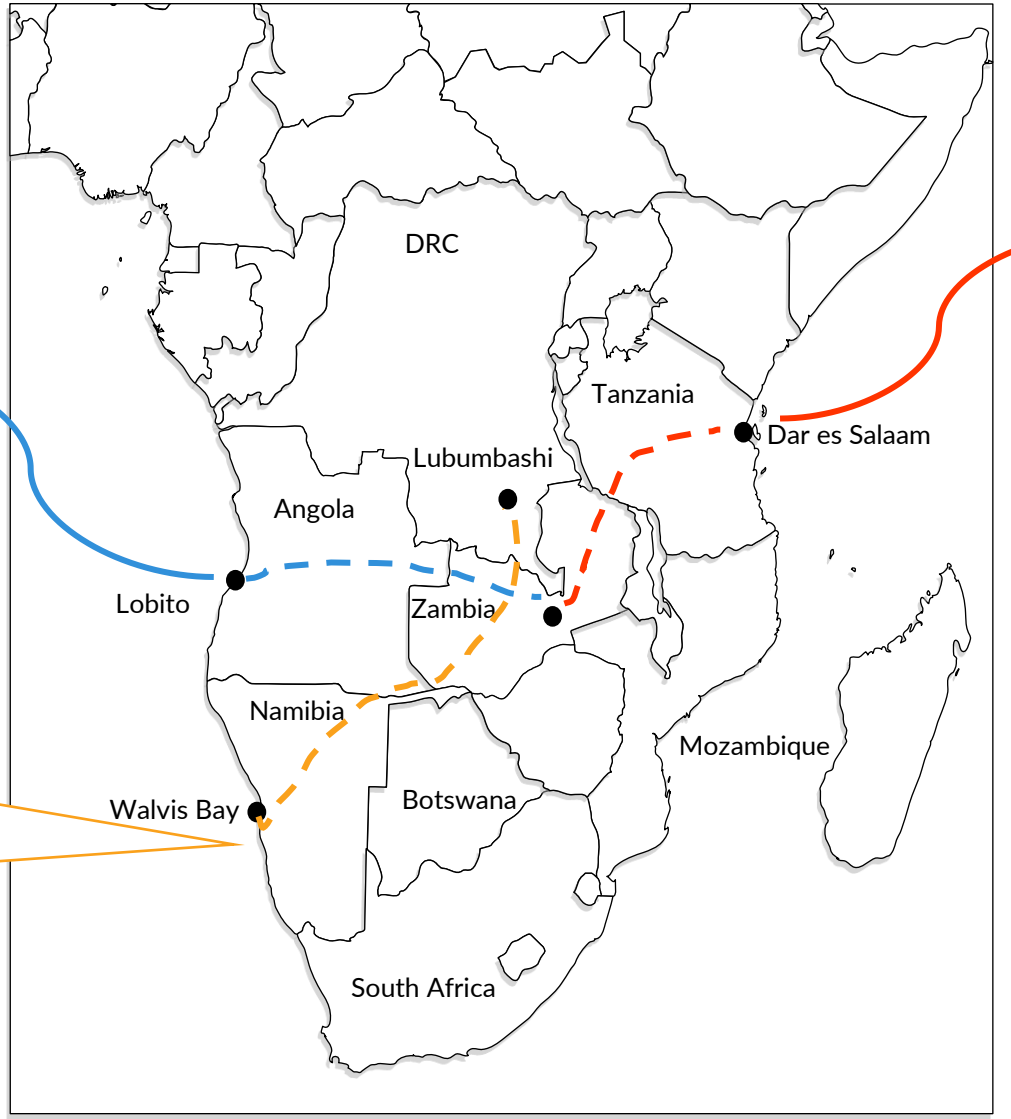
Proposed US and EU backed Lobito Corridor channelling copper and cobalt from the same mineral-rich central Africa belt to the Atlantic coast of Angola.

Key to securing supply chain raw materials with electric vehicles, batteries, green hydrogen demand rising (copper, cobalt, rare earth elements)

Development of alternate routes like *Walvis Bay-Ndola-Lubumbashi Corridor* offers African countries potential leverage in negotiating financing, tariffs and service standards.



Chinese backed Tazara corridor
Dar es Salaam (Tanzania, Indian Ocean) ↔ Kapiri Mposhi (Zambia, copperbelt)

China historically has dominated infrastructure investment in the region.

Diversifies export routes reduce reliance on east-coast ports; framed as "transparent alternative" to Chinese projects

AGENDA

- Current market overview
- Demand analysis
  - Diesel
  - Gasoline
- Demand forecast
- Supply dynamics & forecast
- Namibia-Walvis Bay overview
- **Competitor Analysis – Vitol**
- Biofuels Blending Regulations

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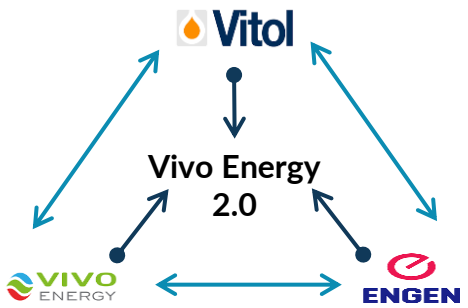
Vitol's strategy in Africa has been acquisition with the goal of growth and owning key parts of the energy value chain..

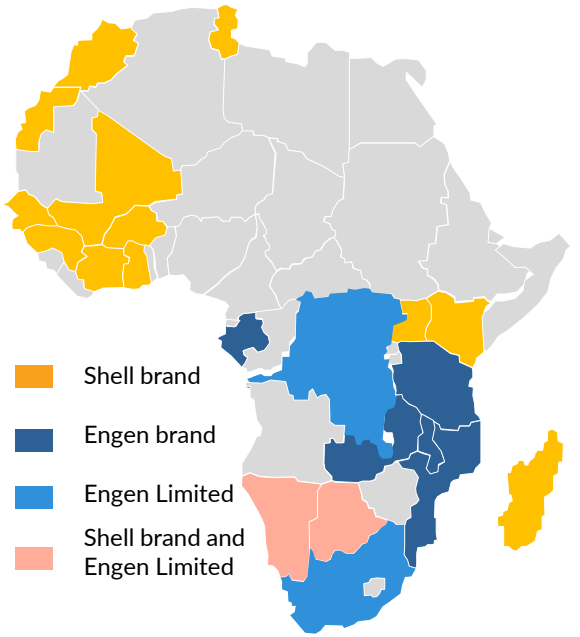
Vopak

Vitol/Vivo Energy Growth Strategy



The bigger picture ambition is to create a new Vivo Energy 2.0

Refinery Status: It will not be restarted as a refinery but repurposed as an import terminal and possibly for gas-to-power (GtP)



Ambition: to be a leading pan-African energy business with a strong growing presence by integrating the businesses and capabilities across the portfolio to unlock upside opportunities

Updates to growth strategy

Legal

- In May 2024, DTIC and Vivo Energy signed the public interest commitments agreed as part of the transfer of ownership of the Engen business in South Africa
- Engen's largest shareholder, Petronas sold its 74% share in Engen, South Africa's largest fuel-distribution network, to Vivo Energy
- The remaining 26% ownership comprises: 21% held by Phembani Group (B-BBEE partner)& 5% allocated to employees via a new Employee Share Ownership Programme

Financial

- R9.85 billion over 5 years to upgrade infrastructure, support local refining through offtake agreements, and fund worker ownership, with scope to increase to R13.85 billion pending feasibility
- Worker Ownership Trust: 5% stake rising to 9%, with dividends and board representation for ~2,100 employees
- ~R100 billion in offtake agreements from local refineries (Astron, Sasol) to support SA fuel production

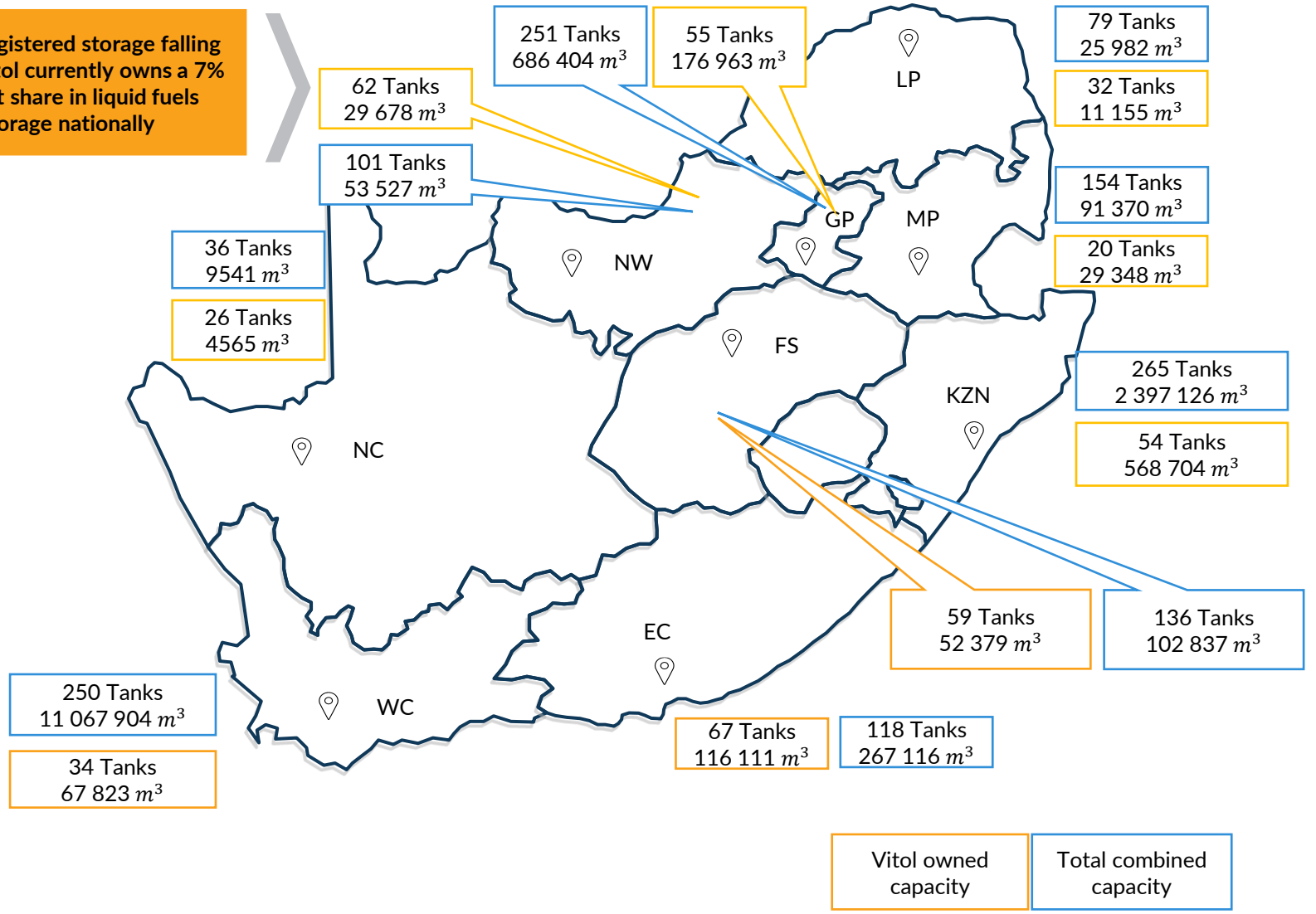
Timelines

- Vitol will implement a new Employee Share Ownership Plan (ESOP) for Engen Petroleum within 6 months of the deal's completion
- Within 5 years, Vitol will increase the ESOP stake to 7%
- Within 7 years, Vitol will increase the ESOP stake to 9%

Sources: The Department of Trade, Industry and Competition | Competition Tribunal

..in light of this, Vitol owns substantial storage nationwide..

Nersa registered storage falling under Vitol currently owns a 7% market share in liquid fuels storage nationally



Africa Strategy

- Expanding fuel storage via VTTI and local partners.
- Integrating trading, storage, and retail operations.
- Partnering with local/state entities to scale regionally.
- Investing in LPG,LNG and biofuels infrastructure.
- Focusing on key hubs across Africa for supply security.

Vitol is positioning itself as a leading, end-to-end fuel supplier across Africa, securing key infrastructure to anchor long-term regional dominance and energy security.

Sources: Nersa

..with infrastructure as leverage, long term commitments such as gas-to-power are possible..

Vopak

Strategy:

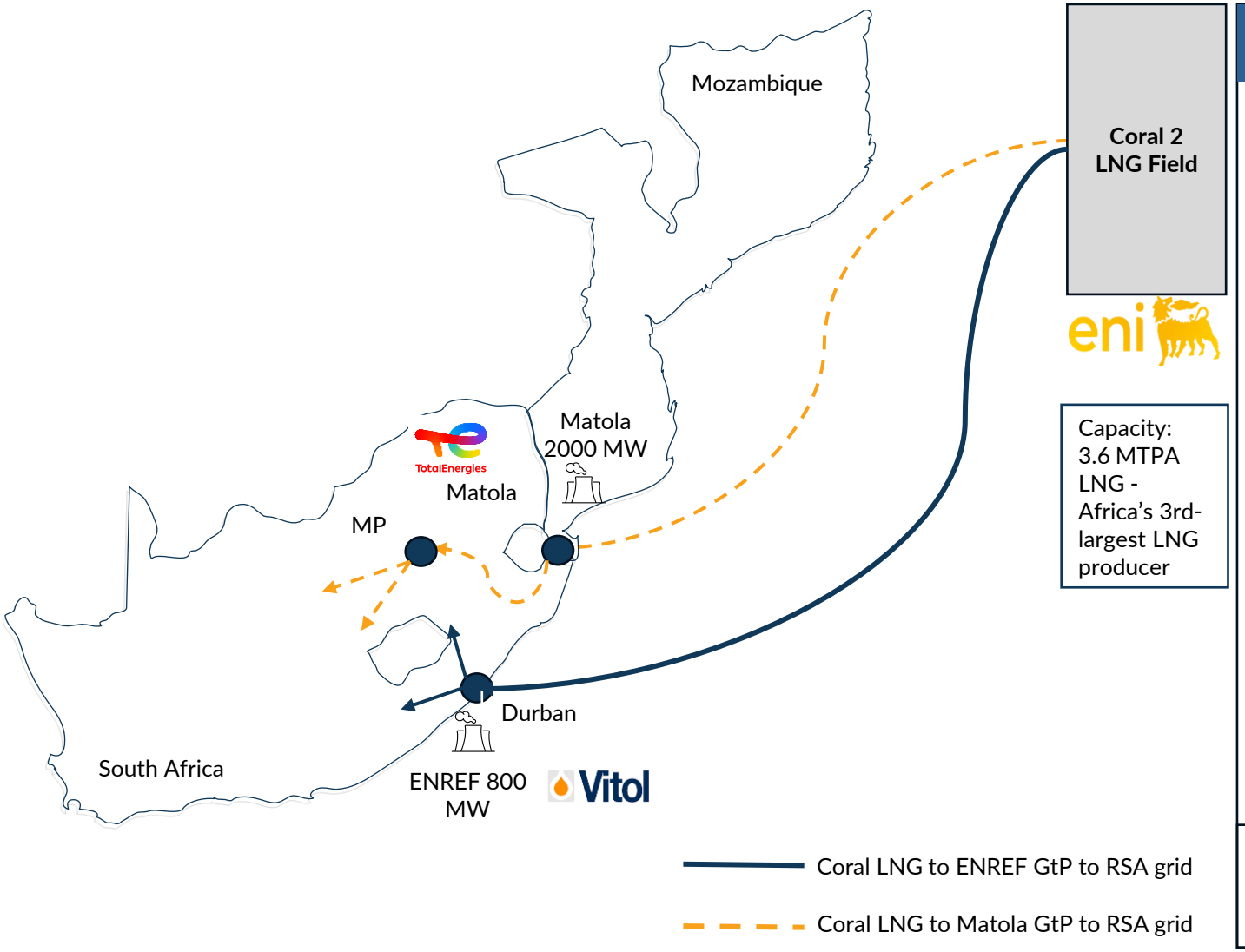
Vitol and ENH formed ENH Energy Trading in 2019 to market LNG, LPG, and condensate

Infrastructure leverage:

Vitol has access through key infrastructure for GtP storage and port terminals in Durban

South African Policy Drivers.

The IRP and Gas-to-Power IPP programs encourage LNG imports and private-sector gas-fired generation to diversify the energy mix.



Key Insights

- Vitol Gas to Power either through ENREF or Matola aligns with policy
- Vitol entrenching influence in Africa's energy transition while securing long-term commercial advantage.

Vitol's play turns storage into leverage, competitors that don't align with the gas-to-power shift risk being sidelined in the next phase of Africa's energy market.

**Further work will be undertaken to gain deeper insights into Vitol's positioning and initiatives in the liquid fuels space.*

..while Vitol are well positioned to realize gas to power projects inline with South African policy aims.

Vopak

End to end gas to power value chain

Global LNG Procurement

- Vitol and ENH formed ENH Energy Trading in 2019 to market LNG, LPG, and condensate.
- Eni's Coral South FLNG began exports in 2022, hitting 5 Mt by 2024.
- Eni approved Coral North (Coral 2) in 2025 to expand Mozambique's LNG capacity.

Vitol well positioned to acquire LNG

Import Terminal (FSRU or Onshore)

- Vitol/Engen operates at Durban's Island View fuel hub with long-term leases (~25 yrs).
- New SBM and CEF/third-party access will affect terminal capacity and land use.

Acquired downstream assets enable LNG import

Gas Transmission (Cryogenic Pipeline)

- Cryogenic pipeline from terminal to power plant
- Requires new right-of-way infrastructure, not shared with oil

Vitol can leverage existing pipelines

ENREF Power Plant /Matola

- 800 MW Enref capacity using high-efficiency turbines
- 2,000 MW combined-cycle gas-to-power plant at Matola
- O&M localisation with Eskom or private contractors
- EPC led by ACWA or similar partner

ENREF/Matola GtP opportunities

Grid Integration (Eskom or Private)

- Connects to high-voltage transmission at ENREF/Matola
- Power sold via PPA or IPP programme, aligned to IRP
- May have opportunity to bid into eThekwini GtP procurement

South Africa open to importing GtP electricity

By embedding in key African energy assets, Vitol positions itself at the center of long-term energy supply and infrastructure control.

Sources: 360 Mozambique

AGENDA

- Current market overview
- Demand analysis
  - Diesel
  - Gasoline
- Demand forecast
- Supply dynamics & forecast
- Namibia-Walvis Bay overview
- Competitor Analysis – Vitol
- **Biofuels Blending Regulations**

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South African decarbonization policy has regulated a reduction in fossil fuels through biofuels blending mandates..

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Regulatory framework

Implications

Blending:

Regulations Regarding the Mandatory Blending of Bio-fuels with Petrol and Diesel (GN R.671 of 2012)

- Licensing and quality assurance
- Diesel: minimum 5% biodiesel (B5)
  Petrol: 2–10% ethanol (E2–E10)
- Into operation on a date determined by the Minister by notice in the Gazette

Guaranteed biofuel quota and blending infrastructure

Volumes & Feedstock:

South African Biofuels Regulatory Framework and National Biofuels Feedstock Protocol (GN 116 of 2020))

- Targeted biofuels penetration is 4.5% v/v of the national fuel pool
  2% expected to come from first generation biofuels technologies
- Feedstock Protocol component : Prioritizes rain-fed crops, under-utilized land, and forbids certain food-critical crops
- Cost recovery mechanism for blending of biofuels

Increased volumes of liquid fuels nationally
Reduced import demand

Pricing:

Biofuels Pricing Regulations (GN R6499 of 2025)

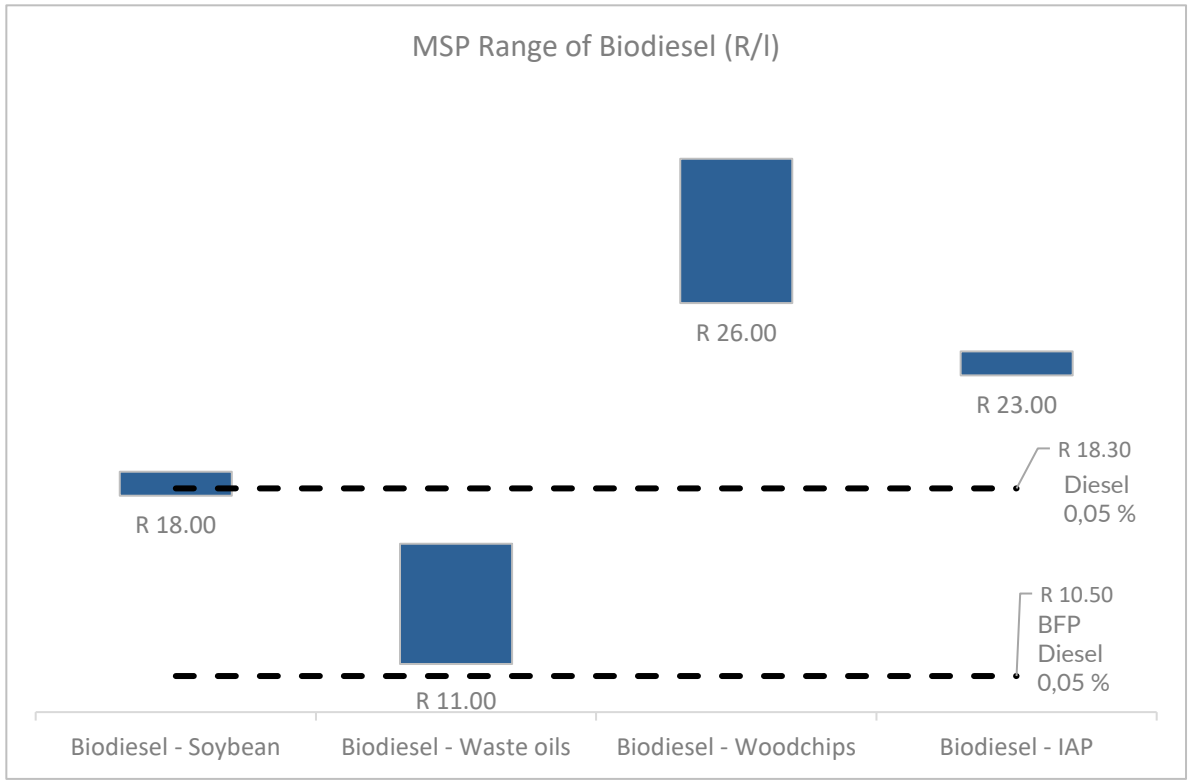
- Regulated Biofuels Price comprises of:

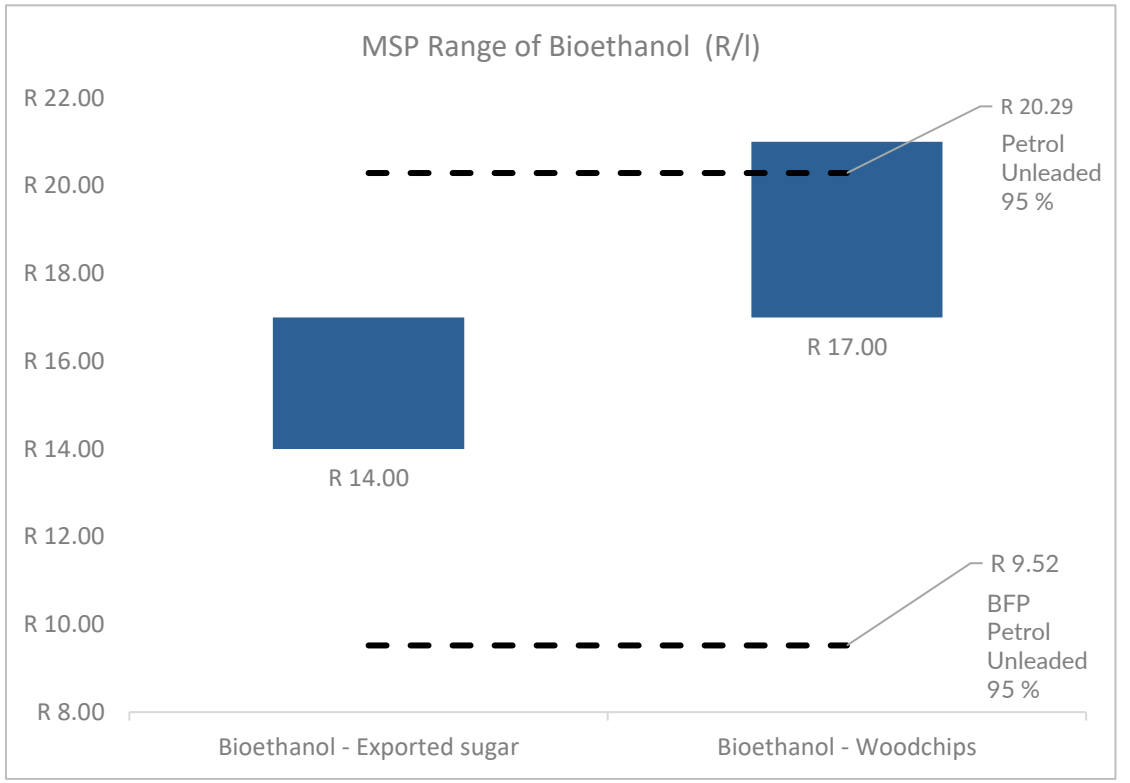
  (a) the import parity pricing for unleaded petrol 95
  (b) the magisterial district zone differential and
  (c) secondary storage costs.

  Monthly Revision of the Regulated Biofuels Price

Cost neutrality for refineries and blenders

..however pricing economics have served as a roadblock to biofuel blending regulation adoption..





- Biodiesel produced from waste oils is the most cost-competitive, with estimated prices of R11–R16 per litre, approaching those of conventional diesel. In comparison, biodiesel from wood chips or invasive plants is significantly more expensive, making waste oils the most practical option.
- Pricing regulations necessary to ensure feasibility.

- Two bioethanol pathways were considered: Exported sugar (traditional fermentation) – R13–R14/litre, Woodchips (cellulosic) – R17–R21/litre
- Both are above the current petrol price (R9,52/litre), indicating pricing support will be needed for competitiveness.

AIA Modelling, MSP: Minimum selling price, BFP (Basic Fuel Price): As of November 2025

..the 2025 Biofuels pricing regulations aim to de-risk biofuel blending and enable implementation..

Biofuels Pricing Regulations pricing model

1. Import parity pricing for petrol 95 (reflecting the international cost of fuel)
2. Magisterial district zone differential (adjusts for local distribution/logistics costs)
3. Secondary storage costs (covers storage/handling infrastructure)

Infrastructure costs

Global fuel markets

Monthly Revisions:

Prices are adjusted monthly to reflect changes in global fuel markets and local logistics, keeping the biofuel price current and aligned with actual costs.

Local logistics

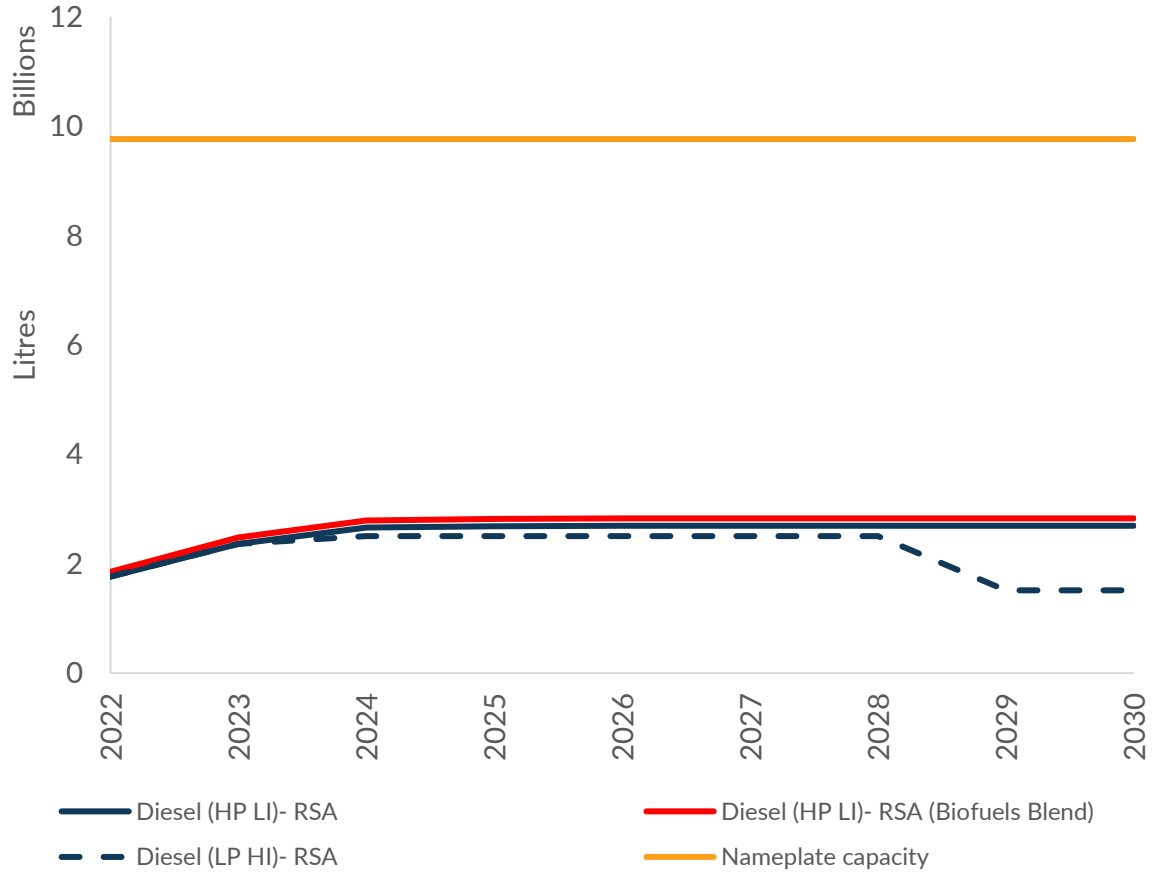
Key Outcomes

- Market-aligned: Prices track international fuel prices, so biofuel is competitive and predictable.
- Regional equity: Zone differentials account for distribution cost variations, so stakeholders in remote areas aren't unfairly penalized.
- Investment certainty: Predictable pricing encourages investment in biofuel production and infrastructure.
- Cost transparency: Helps retailers, traders, and policymakers understand cost components and plan accordingly.

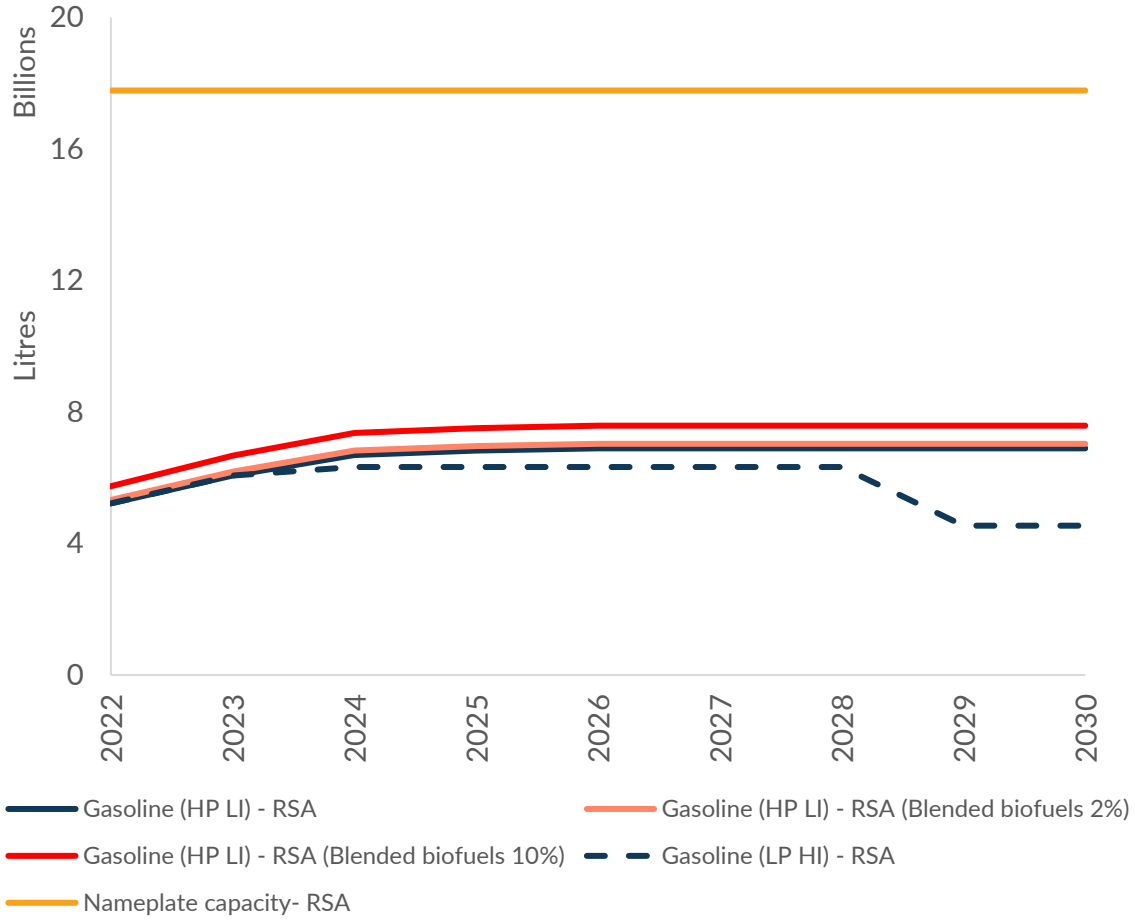
Increased feasibility of biofuels blending regulations coming into play.

..while the implications of blending regulations coming into play includes a change in the South African supply landscape.

Diesel production with blended biofuels (indicative)



Gasoline production with blended biofuels (indicative)



Blending regulation lean the South African market closer towards a low import scenario

Source: DMRE, AIA modelling HP – High production; LP – Low production; HI – High import; LI – Low import

Thank you